

# **CSA 2.0**

## **Advanced design mode (Tables & RDS)**

### **User guide**

DSm, 01/06/2024

Ver. 1.43

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# 1 Introduction

## 1.1 Design methods

Structural Engineering design verifications and optimisation can be performed using two different methods:

### A. Tables

- Introduced in CSA in 2019 (the original method),
- The design is based on pre-calculated load tables,

#### Positives

- The method optimises the design by suggesting the most cost-effective profiles for given geometry and loads,
- The method is very fast.

#### Negatives

- Some structures are not supported by Tables, when they do not comply with assumptions inevitably made during load tables' generation (see section 1.3),
- Some structures are conservative when designed using Tables due to limitations of the method (see section 1.4).

### B. RDS

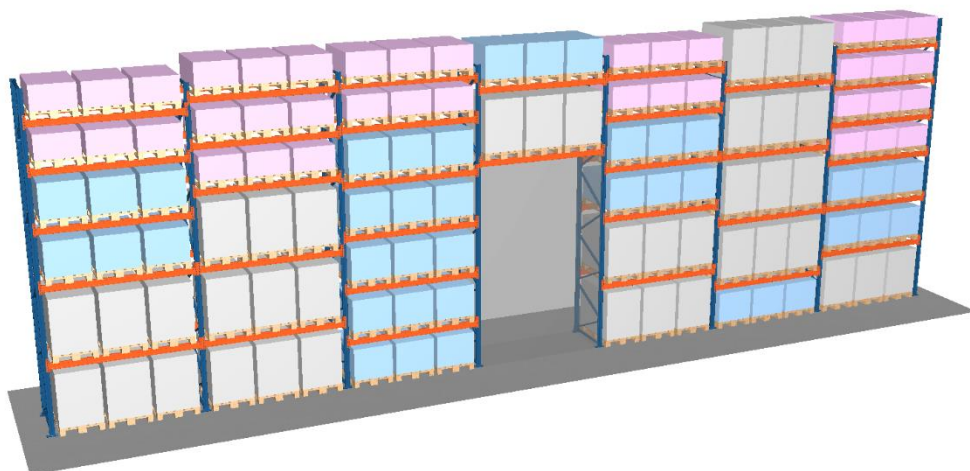
- Introduced in CSA in May 2024,
- The design is executed using RDS (Rack Design Software). RDS is Nedcon's own software for advanced structural engineering design of racking structures. It is based on an industry-leading commercial package RFEM by Dlubal GmbH using the finite element method (FEM).

#### Positives

- The method can analyse almost any structure. It can analyse structures not supported by Tables (see section 1.3),
- The method is always very precise, because actual geometry and loads are modelled. This eliminates the conservative design of some structures by Tables (see section 1.4).

#### Negatives

- The method only verifies profiles assigned either by the user or by the Tables. It can't suggest optimised profiles.
- The analysis can take some time (e.g. 2-3 minutes). On the other hand, this time has to be compared with a human structural engineer working on the project for half a day or so.



## 1.2 Design modes

The CSA RDS Integration introduces the possibility to choose one of the two design modes:

### I. Basic design mode

- using Tables,
- corresponds to CSA operation from 2019 to May 2024.

### II. Advanced design mode

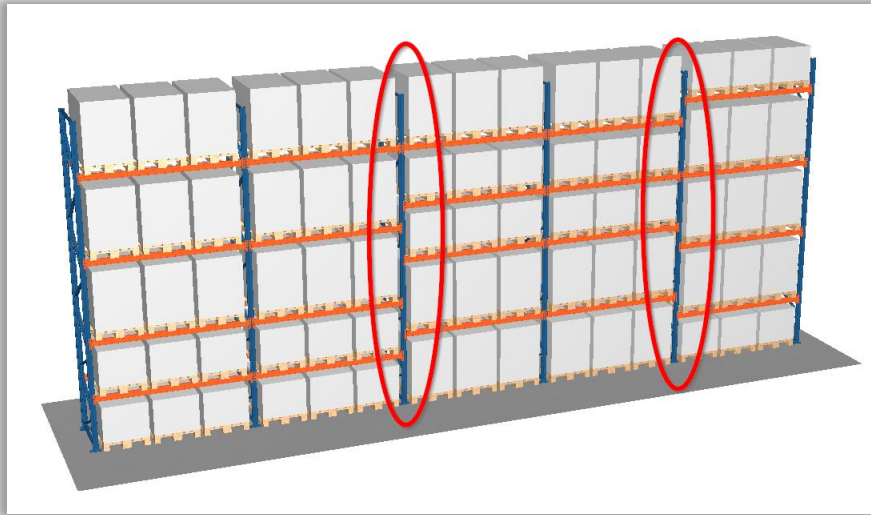
- Using Tables & RDS,
- The user can choose either of the methods or even both of them.

### 1.3 Structures not supported by Tables but supported by RDS

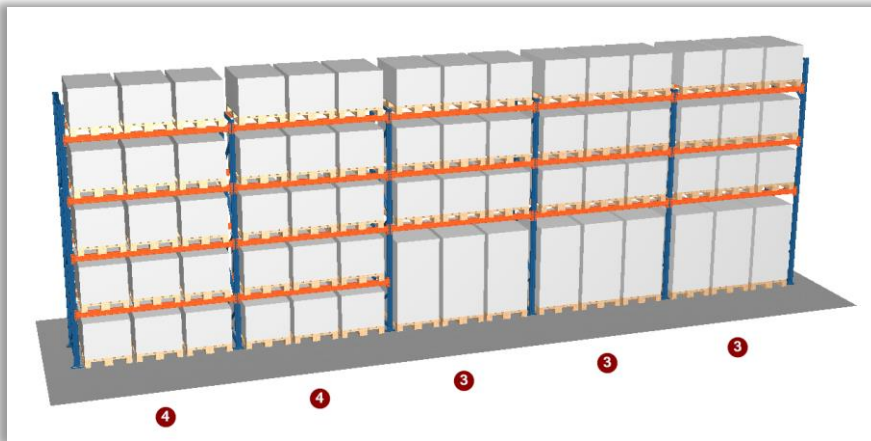
Some structures are not supported by Tables, when they do not comply with assumptions inevitably made during load tables' generation. However, they can be analysed using RDS.

Examples of such rack configurations are:

**a. Rows with irregular beam levels** – beams in adjoining bays are not aligned,

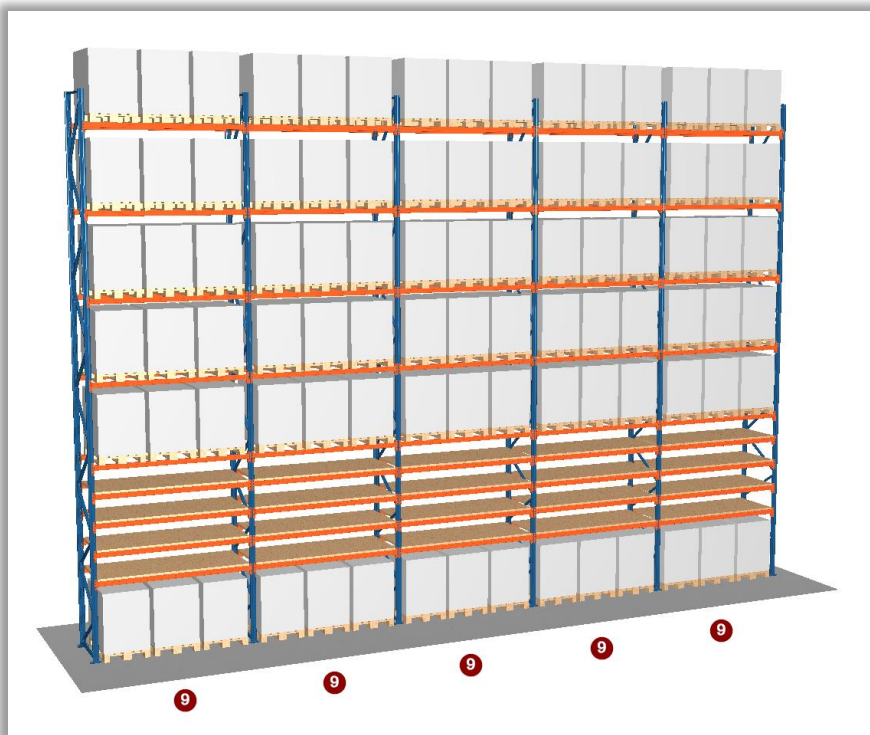


**b. Rows with varying no. of beam levels per bay** – e.g. 4 levels and 3 levels in the same row,



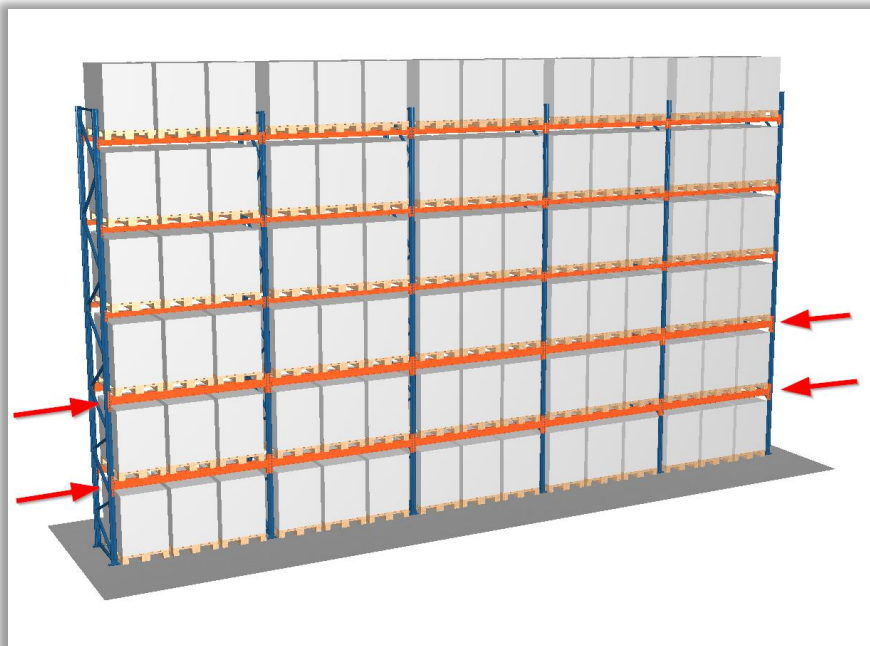
### c. Bays with more than 7 beam levels

Design using Tables is limited to 7 beam levels, design using RDS supports up to 15 beam levels.



### d. Bays with varying beam profiles – important for optimisation of larger projects,

The user may need to have e.g. stronger profiles at the bottom of a bay and lighter profiles on remaining levels.

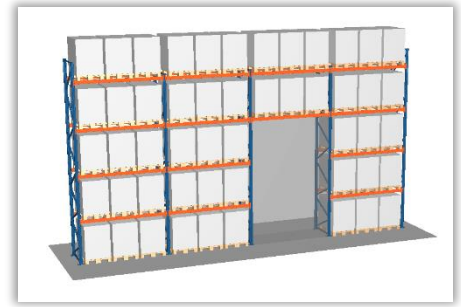


### e. Bays with custom beam configurations

Concerns custom combinations of beams and hook-in connectors created by the user with the help of the new Beam Editor. They are possible only when using RDS design.

**f. Rows with tunnels** - not complying with table limitations:

- rows with less than 5 bays - OR - tunnel frequency less than 1:5
- rows with two tunnels next to each other
- rows are limited to 1 beam type outside tunnel (varying beams in end sections are not possible)



**g. Rows with P&Ds** – not complying with table limitations,

- rows below min. equivalent imperfection => not possible to design
- rows with P&Ds (as extra storage) combined with Tunnels
- rows are limited to 1 beam type (varying beams in end sections are not possible)



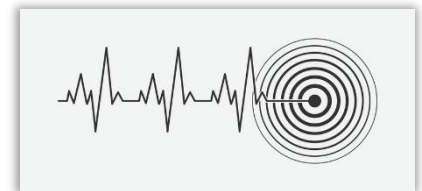
**h. Rows with P&Ds and with Tunnels**

Combination of P&Ds and Tunnels in one row is possible only when using RDS design.

**i. Seismic projects** – other than Germany, Zones 0 & 1.

CSA Tables design supports seismic projects only in Germany and only in Zones 0 and 1.

RDS design on the other hand supports seismic projects in 12 European countries incl. Germany, Switzerland and Austria.

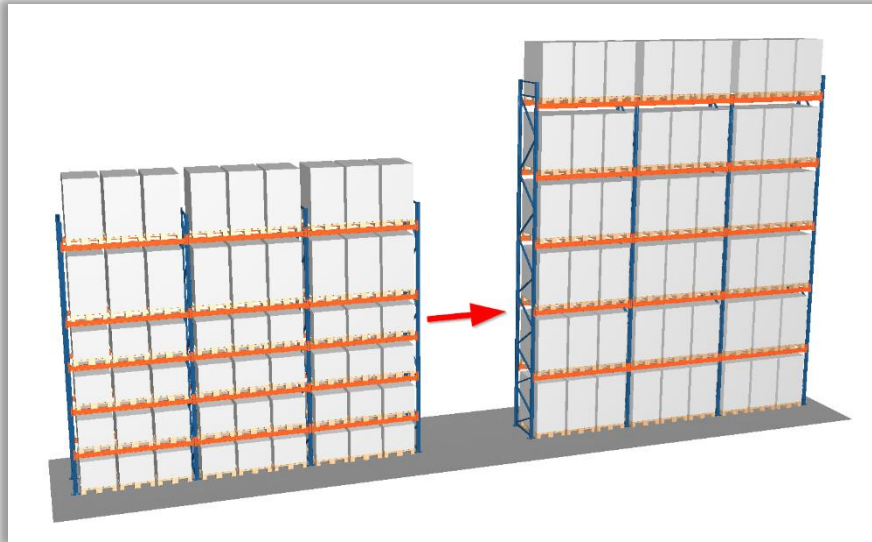


## 1.4 Structures conservative when using Tables but precise when using RDS

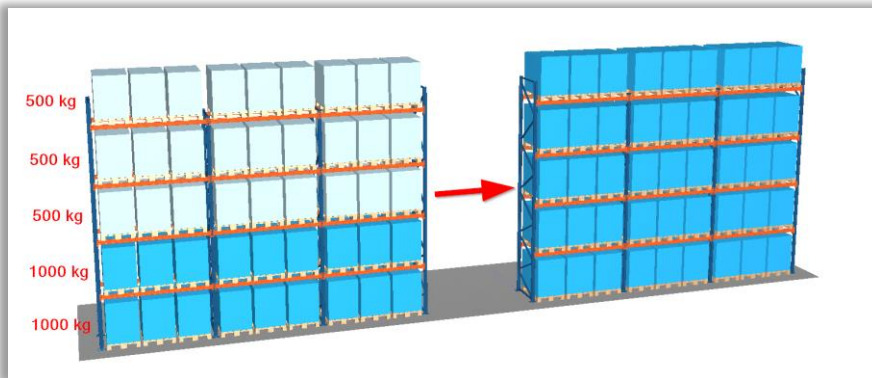
Some structures are conservative when designed using Tables due to limitations of the method. However, they are analysed precisely using RDS.

Examples of such rack configurations are:

**a. Bays with varying compartment heights** – designed using Tables as if all compartments are high,



**b. Bays with varying pallet weights** – designed using Tables as if all pallets are heavy,



**c. Shorter rows** – rows having 4, 3, 2 or 1 bays,

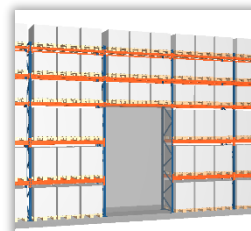
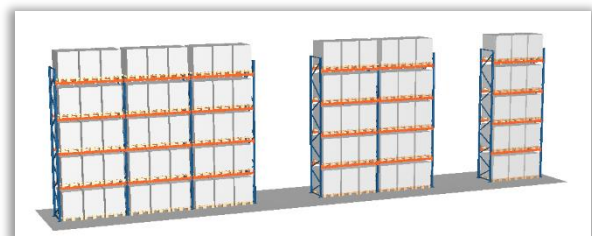
- conservative approach is used (equiv. imperfection)

**d. Rows with Tunnels**

- conservative approach is used (equiv. imperfection)

**e. Rows with P&Ds**

- conservative approach is used (equiv. imperfection)

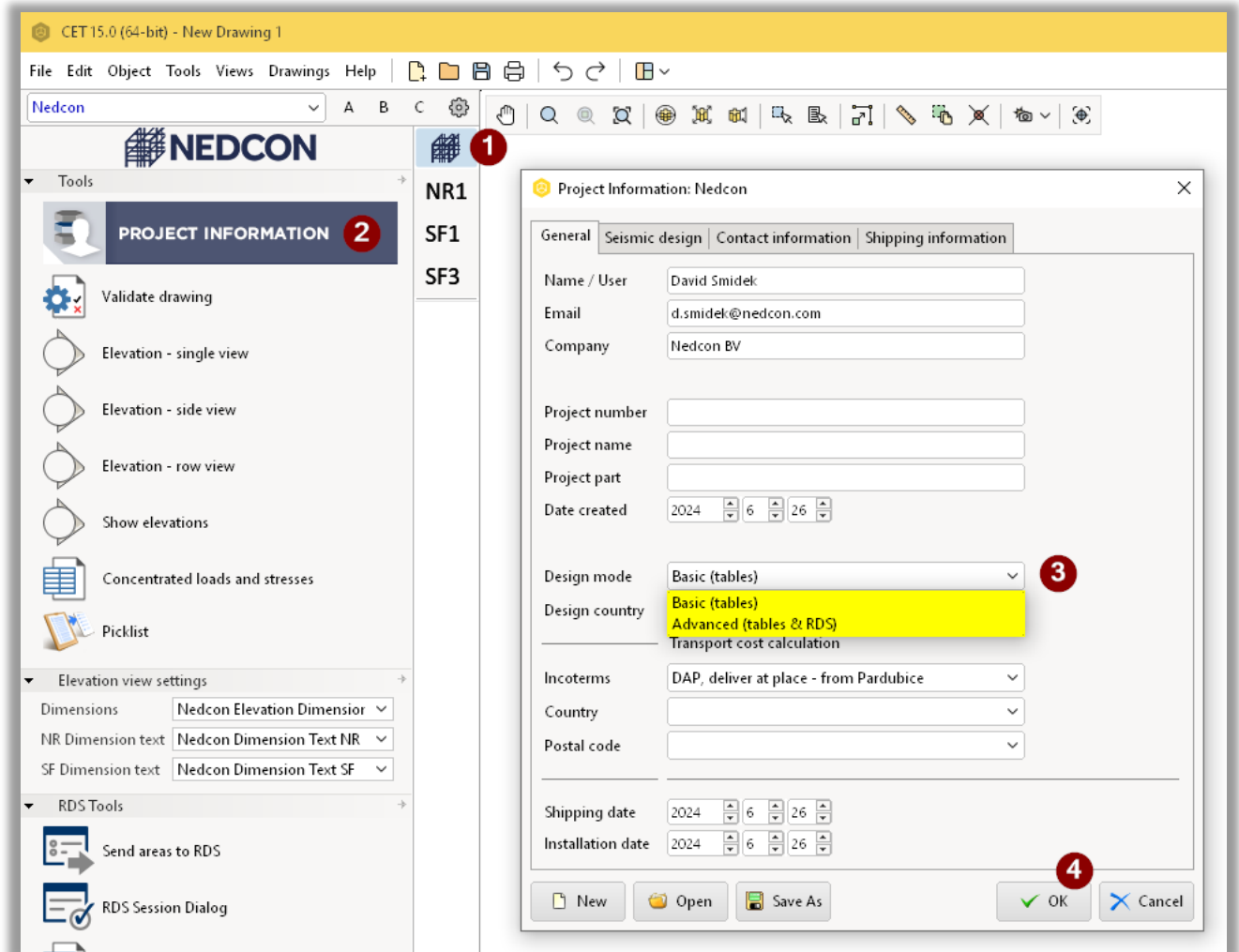


## 2 Design process

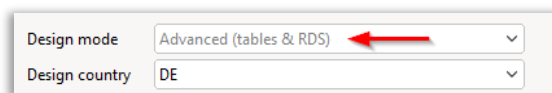
### 2.1 Switching between Basic and Advanced design modes

CSA operates in the Basic design mode (Tables only) as default. If the user wishes to use also the RDS design, then he/she needs to switch to the Advanced design mode (Tables & RDS) :

1. Go to tab **Nedcon** (1) and open **Project Information** (2) in tab **General**,
2. Using pull-down menu **Design mode** select **Advanced (Tables & RDS)** (3),
3. Confirm **OK** (4).



The user can switch between Basic and Advanced design modes until an RDS analysis is executed. The project stays then locked in Advanced mode. It is not possible to switch to Basic mode anymore.



The user can however still perform structural design using either Tables or RDS.

If the user performed any changes in the Advanced mode, which are not supported by the Tables design (e.g. 2 Tunnels next to each other), he/she then needs to modify the structure manually back to a state which is supported by the Tables before running Structural Design using Tables.

## 2.2 Rules for rows to be covered by a single RDS run (Virtual row)

### a. General

RDS is able to analyse only one racking row at a time.

If multiple rack rows are to be analysed by a single RDS run, they need to be identical.



### b. Identical rows

Rows are identical, if they have identical:

- Geometry - all dimensions and quantities are identical,
- Profiles - all profiles are identical,
- Loading - all compartment loads are identical and applied on identical locations.

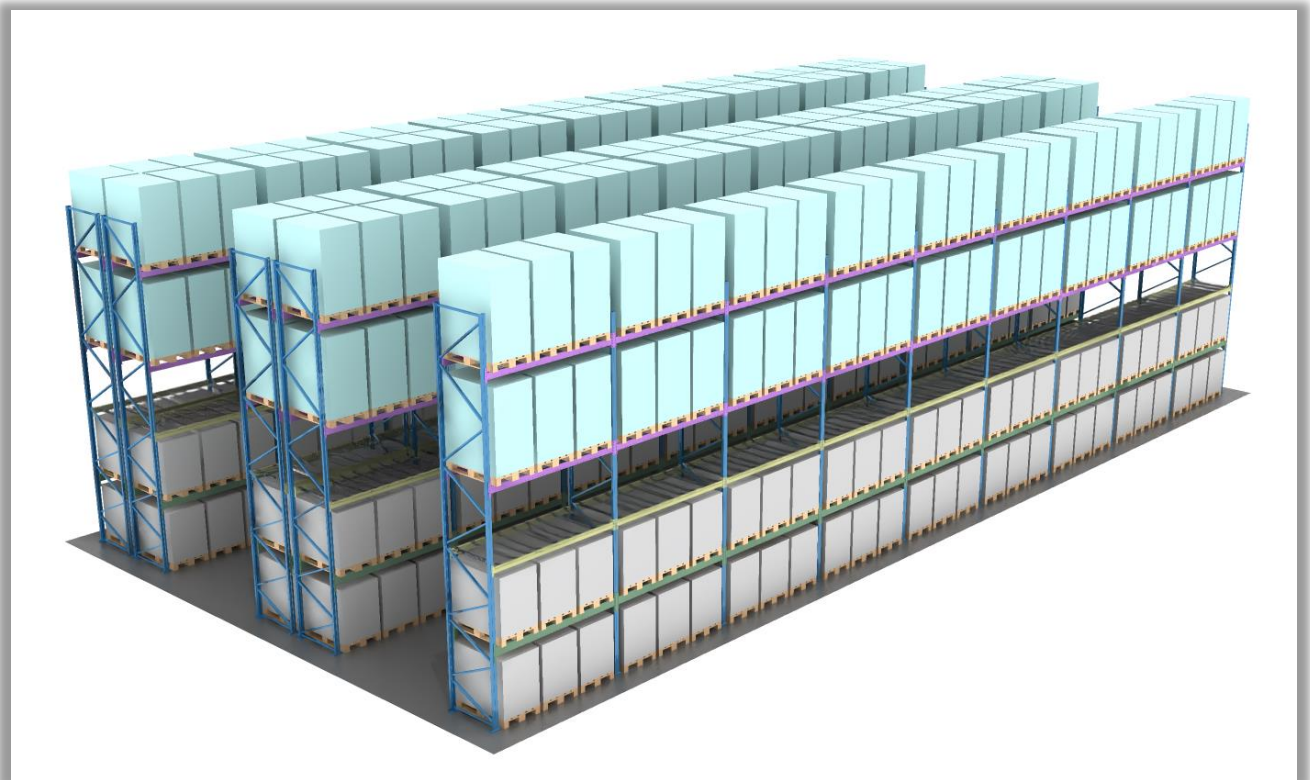


The compartment load consists of the weight of load units and the weight of beam accessories.

Rows are identical irrespective of the fact whether they form single or double rows.

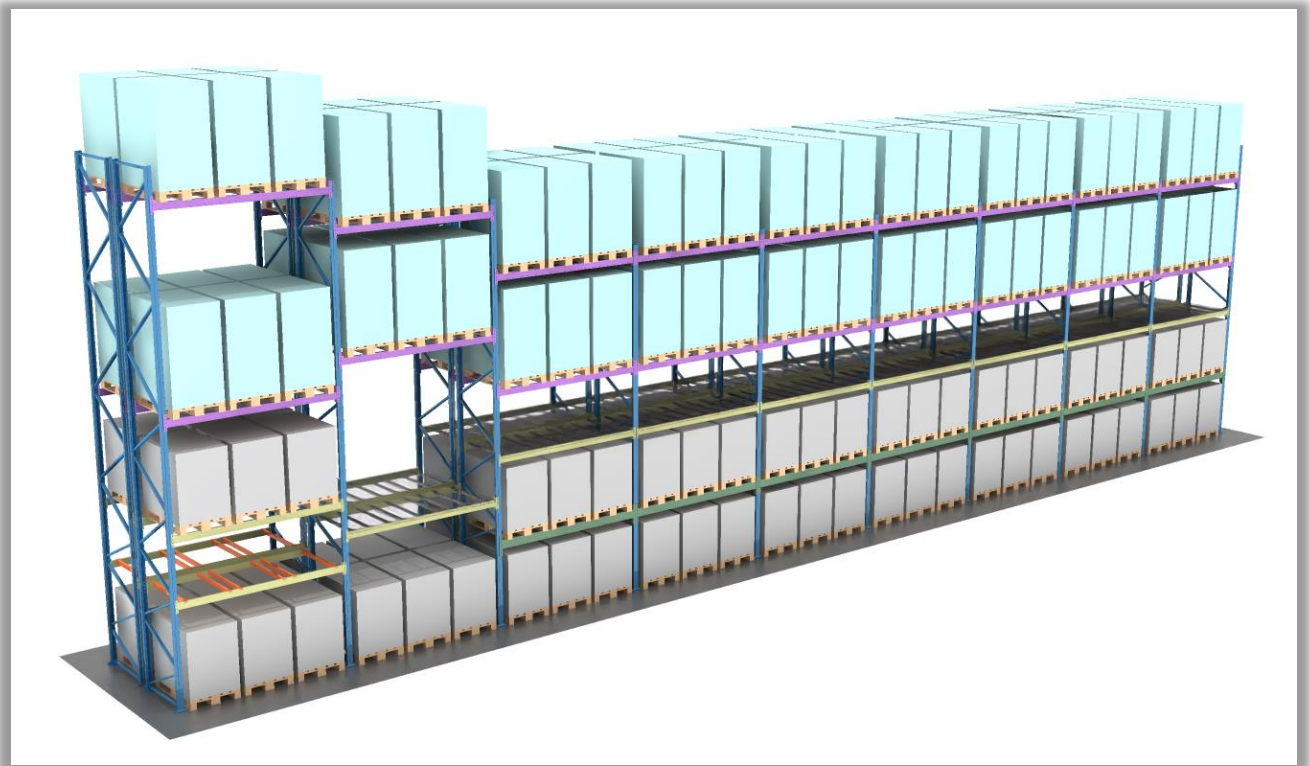
Rows, which are mirror images of each other, are identical as well.

Example – identical rows:



Example – identical mirrored rows:

The front row has the irregular bays on the left-hand side, whereas the rear row on the right hand side. Yet, both rows are identical.



A group of identical rows can be analysed using a single run of RDS.



### c. Varying rows

Varying rows are rows which are not identical.

A group of varying rows has to be split into sub-groups of identical rows. Each sub-group of identical rows has to be analysed using a dedicated single run of RDS. Multiple RDS runs are then executed.



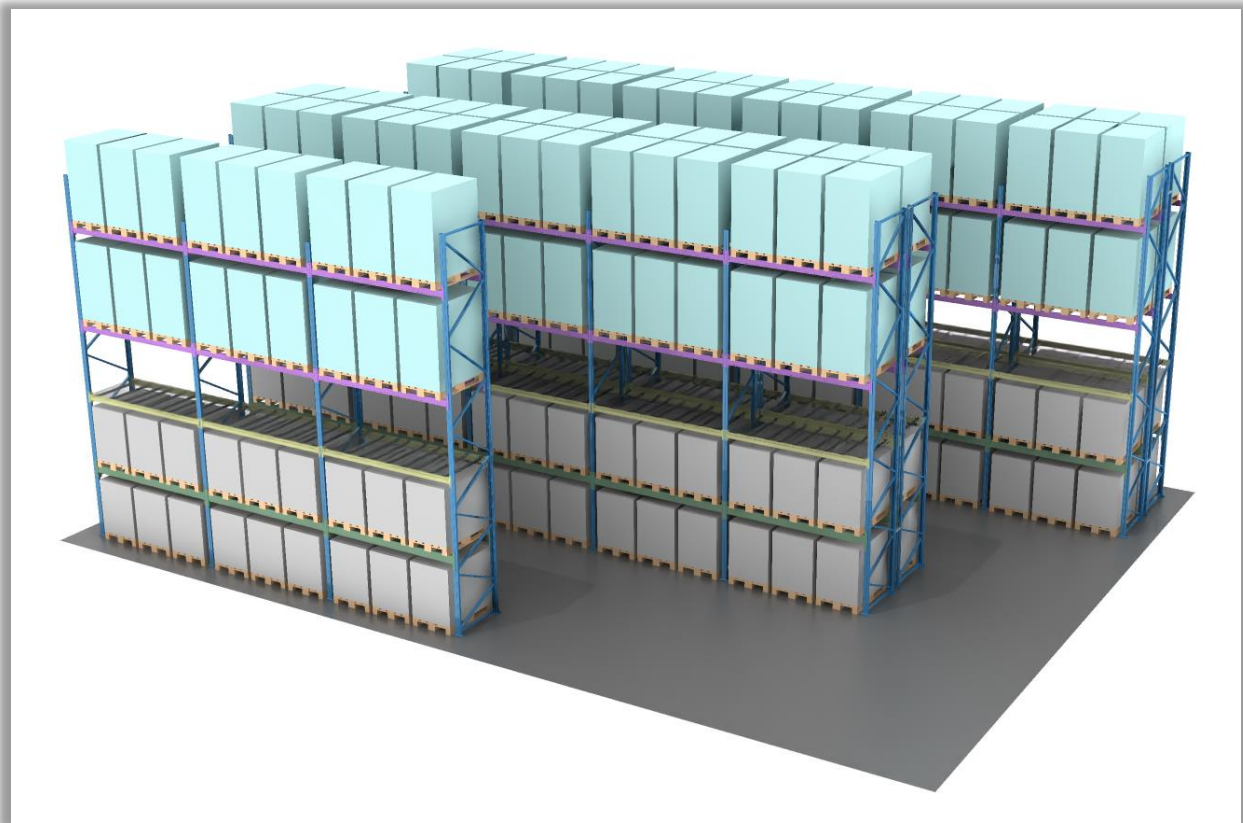
#### d. Varying rows - Special case - Rows with identical bays, varying in number of bays only

In this special case, even varying rows can be covered by a single RDS run. RDS then analyses a fictitious row, which represents all rows of the group. Such row in RDS is then called a **Virtual row**.

The special case concerns a group of rows, which:

- have 3 bays or more,
- vary in number of bays per row,
- but all bays are identical (i.e. all bays have identical geometry, identical profiles per given level and identical compartment loads per given level),
  - Within one bay there may be multiple profiles and multiple compartment loads used, but all bays of the group shall be identical.
  - The end frames may have different frame height than intermediate frames. CSA considers just intermediate frames when determining parameters for RDS analysis.
- do not contain any Pick and Drop stations.

Example - Special case - Rows with identical bays, varying in number of bays only:



A group of rows with identical bays, but varying in number of bays only (special case), can be analysed using a single run of RDS.



RDS then analyses a row, which has a number of bays equal to the longest row in the group. A study shows that such **virtual row** generates the most unfavourable component checks and represents thus all rows in the group.

##### Note 1

Rows with 1 bay behave differently than rows with 3 bays and more. They have to be therefore analysed in a separate RDS run.

## Note 2

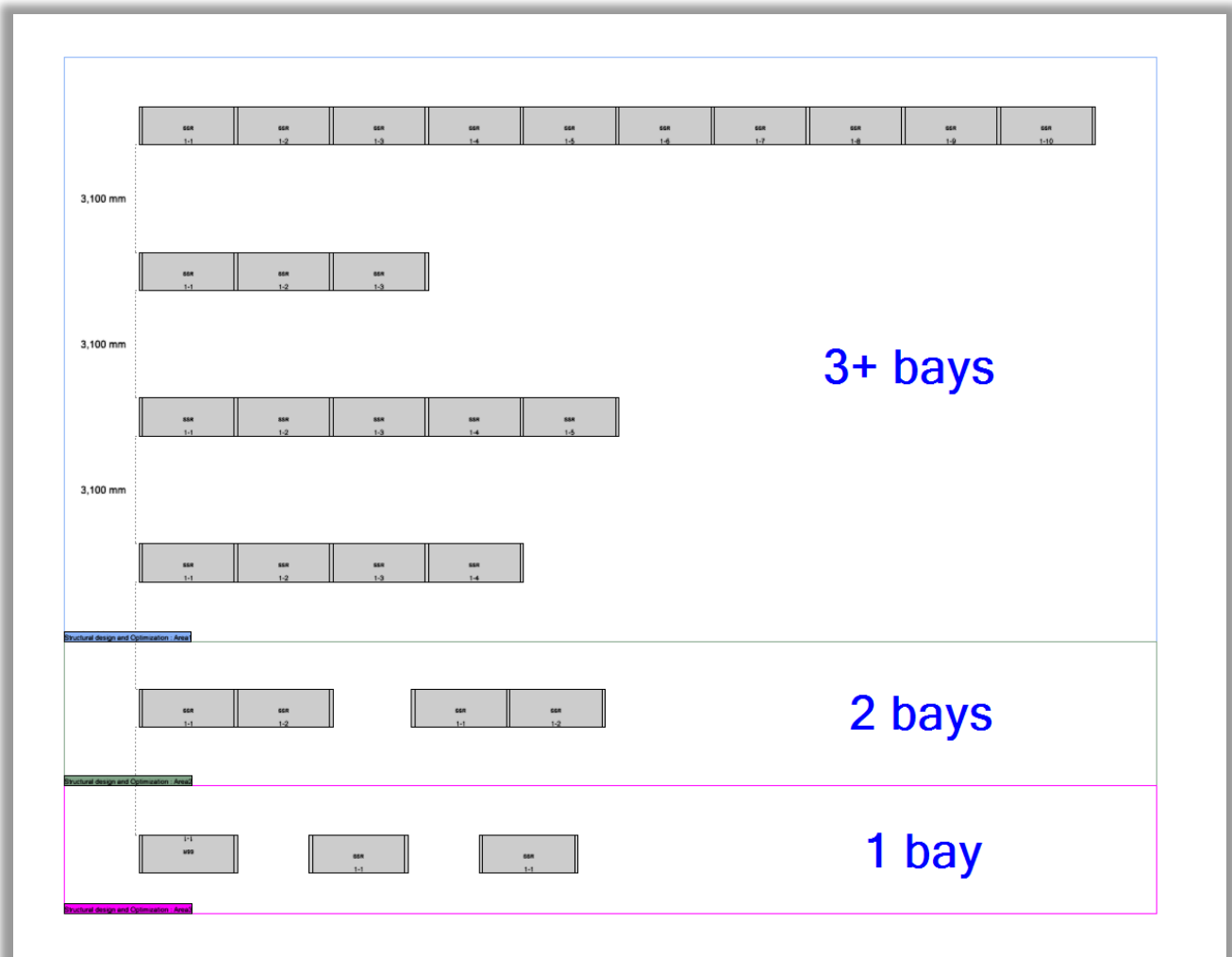
Rows with 2 bays behave differently than rows with 3 bays and more. They have to be therefore analysed in a separate RDS run as well.

## Example

A project contains rows with 1, 2, 3, 4, 5 and 10 bays. Assuming all bays are identical, the project has to be split in 3 groups of rows:

- group of rows having 3 bays and more,
- group of rows having 2 bays,
- group of rows having 1 bay.

Each group of rows can then be analysed using a single run of RDS.

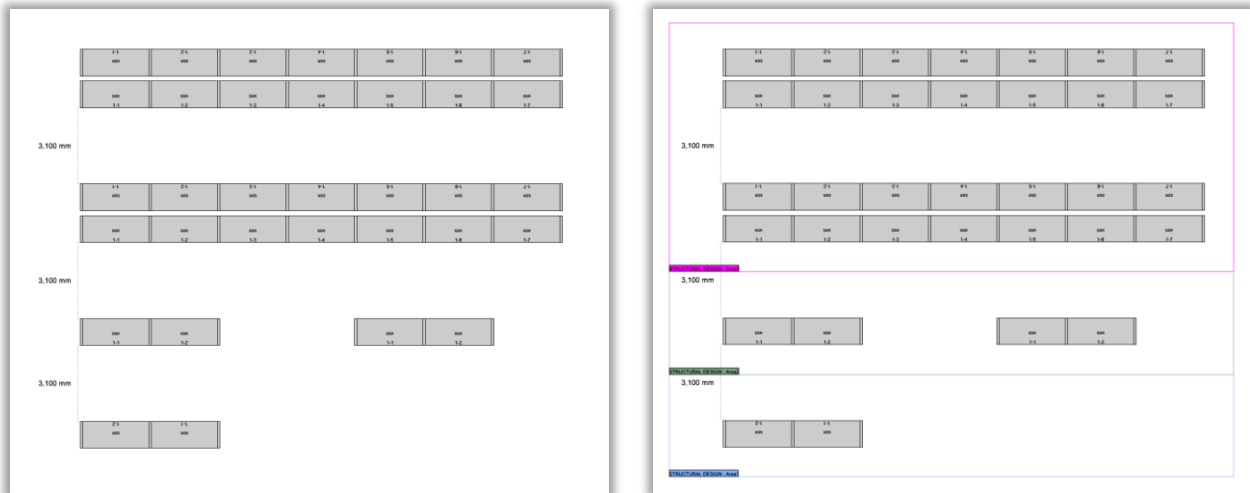


## 2.3 Definition of areas to be analysed (tags)

The user may wish to run the RDS design on:

- All drawing** (all rack rows in a project),
- STRUCTURAL DESIGN tags** – i.e. specified portions of a project (some rack rows).

Example - All drawing vs. three STRUCTURAL DESIGN tags defined:



### a. All drawing

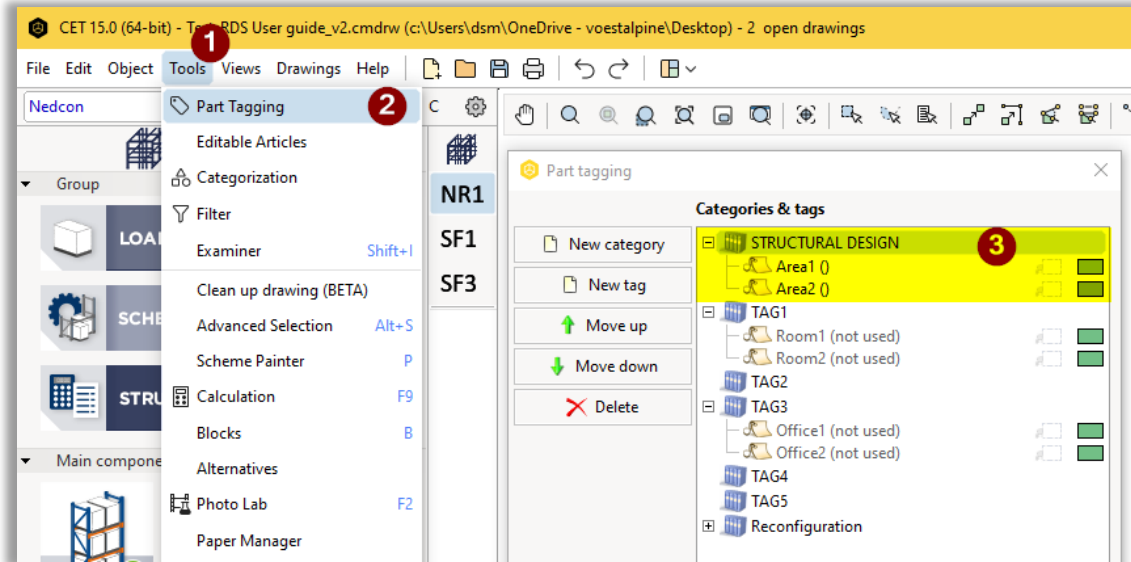
The user does not need to do anything special. If no STRUCTURAL DESIGN tags are defined, the complete drawing is considered in the structural verifications (using either Tables or RDS).

See section 2.2 for Rules for rows to be covered by a single RDS run (Virtual row).

## b. STRUCTURAL DESIGN tags

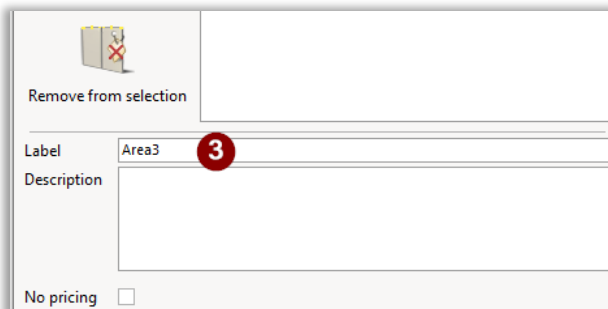
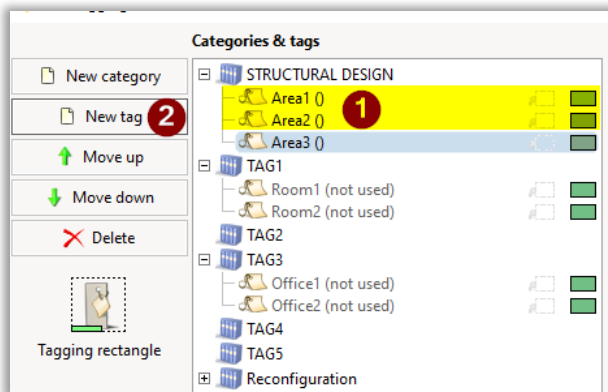
A row or group of rows to be analysed by a single RDS run needs to be marked (tagged) by a STRUCTURAL DESIGN tag:

1. Go to **Tools** (1) and select **Part Tagging** (2),
2. A dialog opens, notice the tag category called **STRUCTURAL DESIGN** (3).



Two empty STRUCTURAL DESIGN tags are pre-defined. More tags in the category can be added and all tags can be renamed:

1. Click on any of the **existing tags** (yellow area) (1),
2. Click on **New tag** (2)
3. Finally, **rename** the newly add tag or any existing tags (3) to your liking.



## Tag names

Note that tag names are the only text information which is transferred from CSA to RDS. If you therefore wish to identify your project or its part (tag) in RDS later, use corresponding, descriptive tag names.

If no tags are defined, the complete project is automatically named as “allDrawingTag” in RDS.

Example – Project names in RDS:

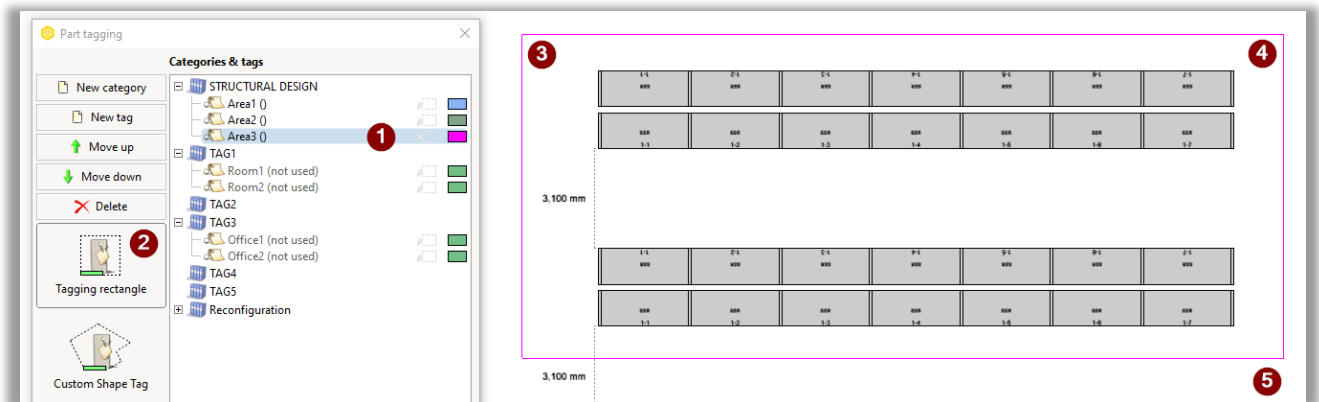
| RDS project ID | External project ID             | System template |
|----------------|---------------------------------|-----------------|
| RDS_24008009   | CSA_allDrawingTag               | MAN NR1         |
| RDS_24007657   | CSA_Area1 - EQ - 12 Switzerland | MAN NR1         |



## Adding tags to a drawing

Tags can be added to a drawing as follows:

1. Select the **desired tag** (1),
2. Click on **Tagging rectangle** (2),
3. Define the **first corner** (3) and the **second corner** (4) of the tag,
4. Then drag the rectangle in perpendicular direction to define the **tag's height** and confirm by last click (5).



## Switching displaying of tags in 2D on and off

The tags are displayed in 2D view automatically when the user is operating in the dialogue **Tools -> Part tagging**.

It is further possible to switch the tags on and off anytime by clicking on the **tabs TAGS** and **Normal** alongside the bottom edge of the screen:

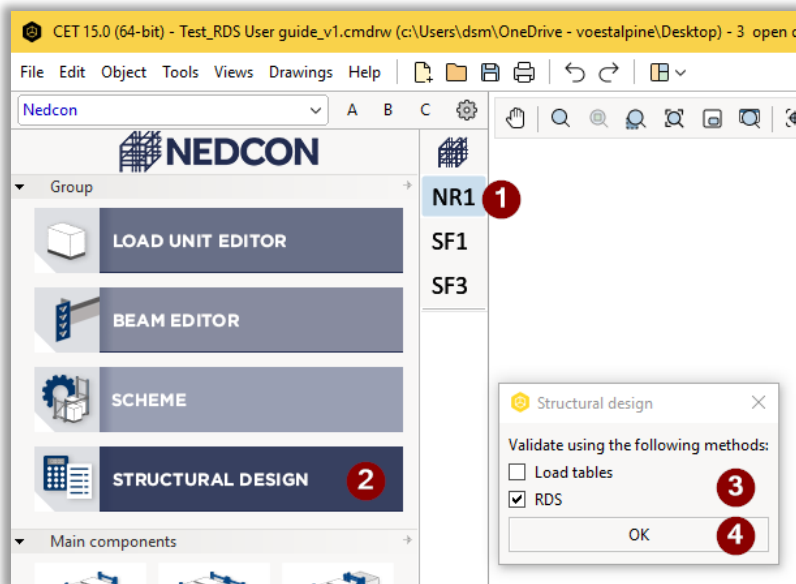
Switching tag display on (TAGS) and off (Normal):



## 2.4 Start structural design and choose analysis method

The process of structural design (verification and optimisation) of pallet racking systems is started as follows:

1. Go to **tab NR1** (1),
2. Click on **STRUCTURAL DESIGN** (2), then dialogue **Structural design** opens,
3. Select **desired analysis method or methods** (3),
4. Confirm **OK** (4).

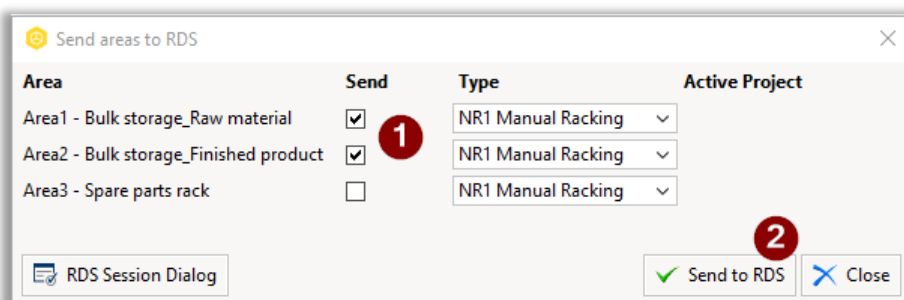


It is possible to design all tags using Load tables or all tags using RDS or some tags using Load tables and some tags using RDS.

## 2.5 Send areas to RDS

The dialogue Send areas to RDS then appears.

1. **Select areas** to be analysed in RDS (1),
2. Click **Send to RDS** (2).



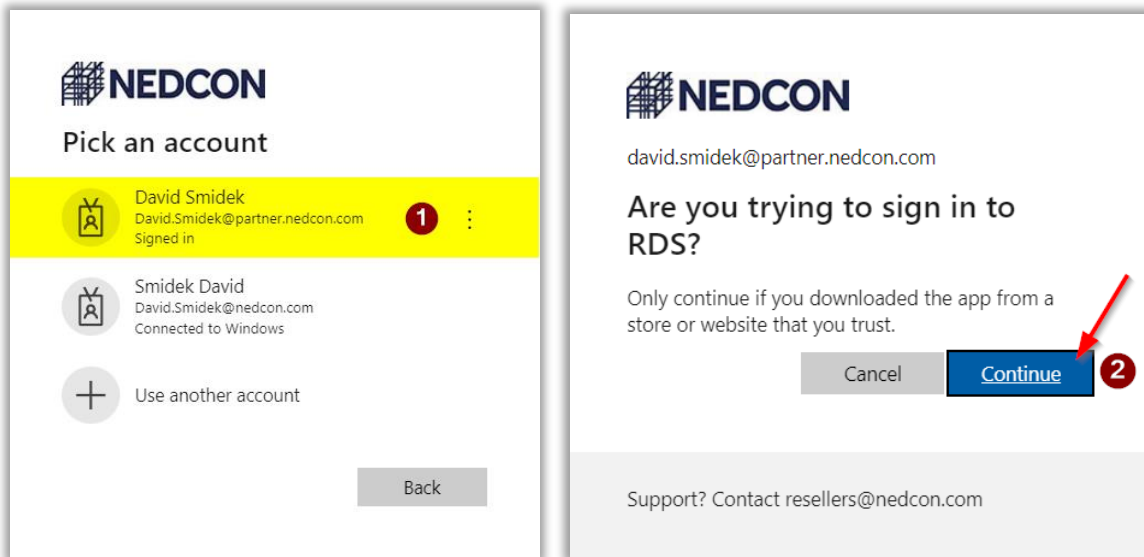
Each selected area will be analysed in a separate RDS run. These RDS runs will be executed concurrently, depending on state of the server hosting RFEM.

## 2.6 Sign in to RDS

Each CSA user needs to have an active Nedcon Partner portal account to be able to analyse structures in RDS.

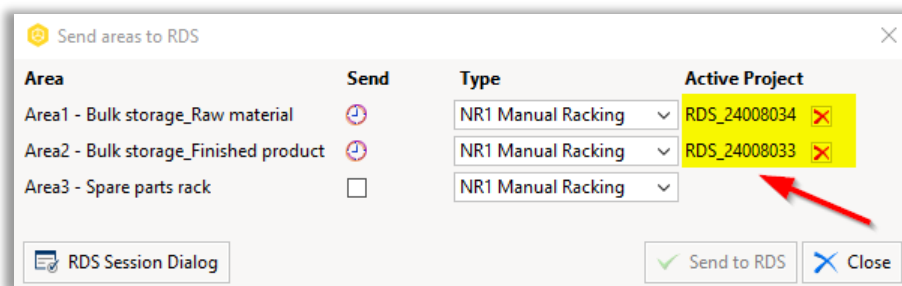
The sign in process is initiated automatically in a default web browser window after sending the first areas to RDS:

1. Select your **Nedcon Partner portal** account (1),
2. Click **Continue** (2),
3. The dialogue disappears and the browser window turns blank.
4. You may **close** the browser window now and **return to CSA**.



## 2.7 Fetching RDS project number

CSA asks RDS to assign a unique, automatically generated RDS project number to each analysis.

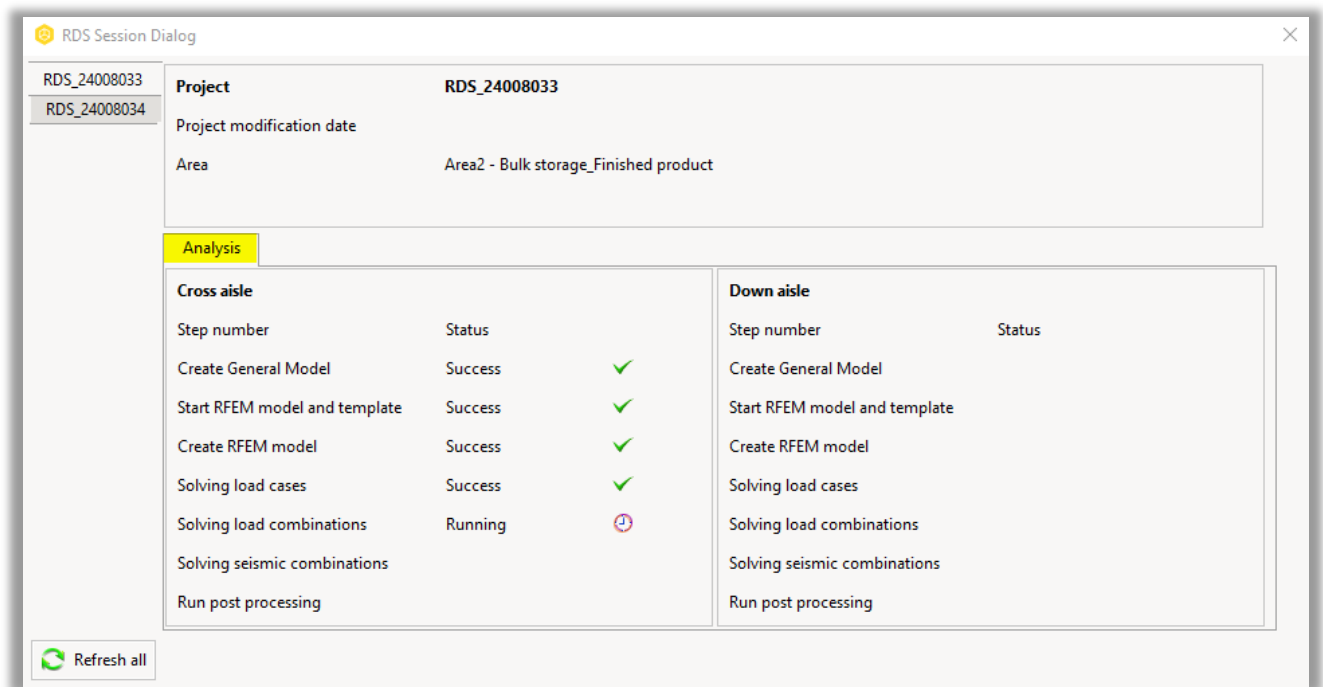


The RDS project number ties the RDS analysis to a specific CSA project or its part (tag).

It can be also used later to identify the CSA analysis in RDS.

## 2.8 RDS Session Dialogue

The application then opens the RDS Session Dialogue. It shows progress of the analysis:



**RDS Session Dialogue**

☐ RDS\_24008033  
☒ RDS\_24008034

**Project** RDS\_24008033  
 Project modification date  
 Area Area2 - Bulk storage\_Finished product

**Analysis**

| Cross aisle                   |         |   | Down aisle                    |        |  |
|-------------------------------|---------|---|-------------------------------|--------|--|
| Step number                   | Status  |   | Step number                   | Status |  |
| Create General Model          | Success | ✓ | Create General Model          |        |  |
| Start RFEM model and template | Success | ✓ | Start RFEM model and template |        |  |
| Create RFEM model             | Success | ✓ | Create RFEM model             |        |  |
| Solving load cases            | Success | ✓ | Solving load cases            |        |  |
| Solving load combinations     | Running | 🕒 | Solving load combinations     |        |  |
| Solving seismic combinations  |         |   | Solving seismic combinations  |        |  |
| Run post processing           |         |   | Run post processing           |        |  |

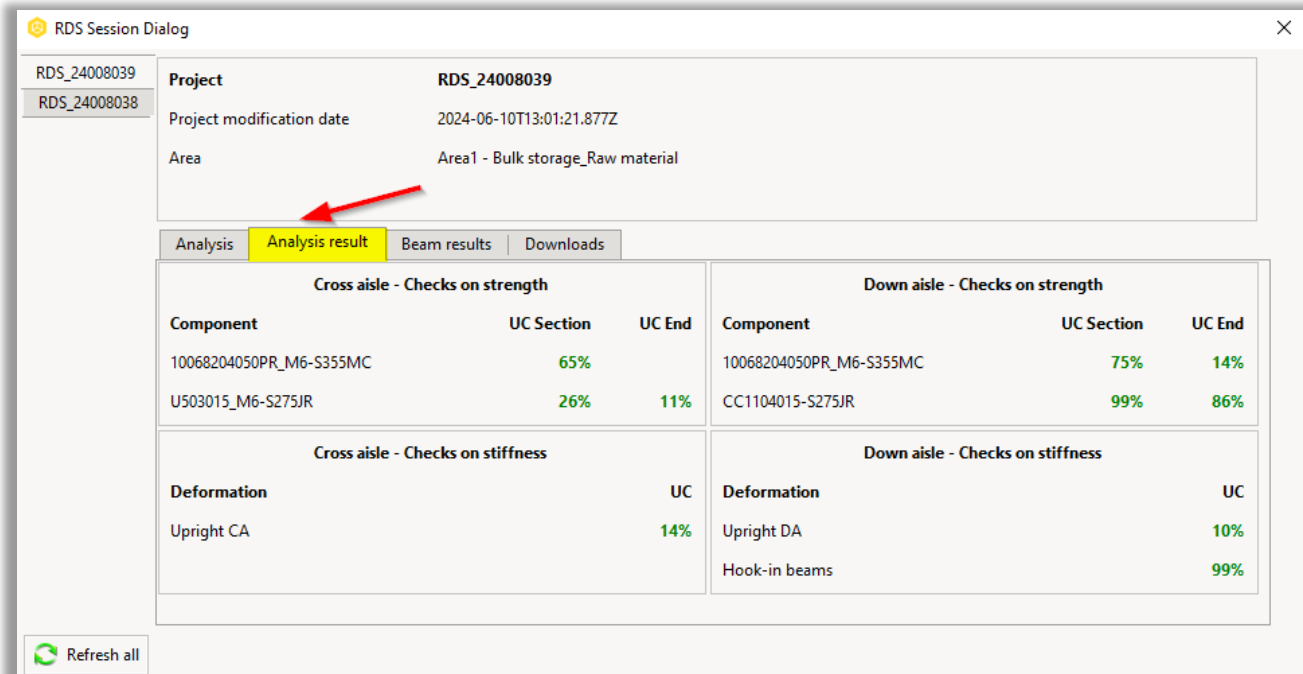
 Refresh all

The analysis can last approximately 1,5 – 3 minutes, depending on its complexity and server load. Multiple RDS runs can be executed concurrently, again depending on the server state.

## 2.9 View analysis results

When finished, the RDS Session Dialogue displays also analysis results.

Click tab **Analysis result**:



**RDS Session Dialogue**

Project: RDS\_24008039  
Project modification date: 2024-06-10T13:01:21.877Z  
Area: Area1 - Bulk storage\_Raw material

Analysis | **Analysis result** | Beam results | Downloads

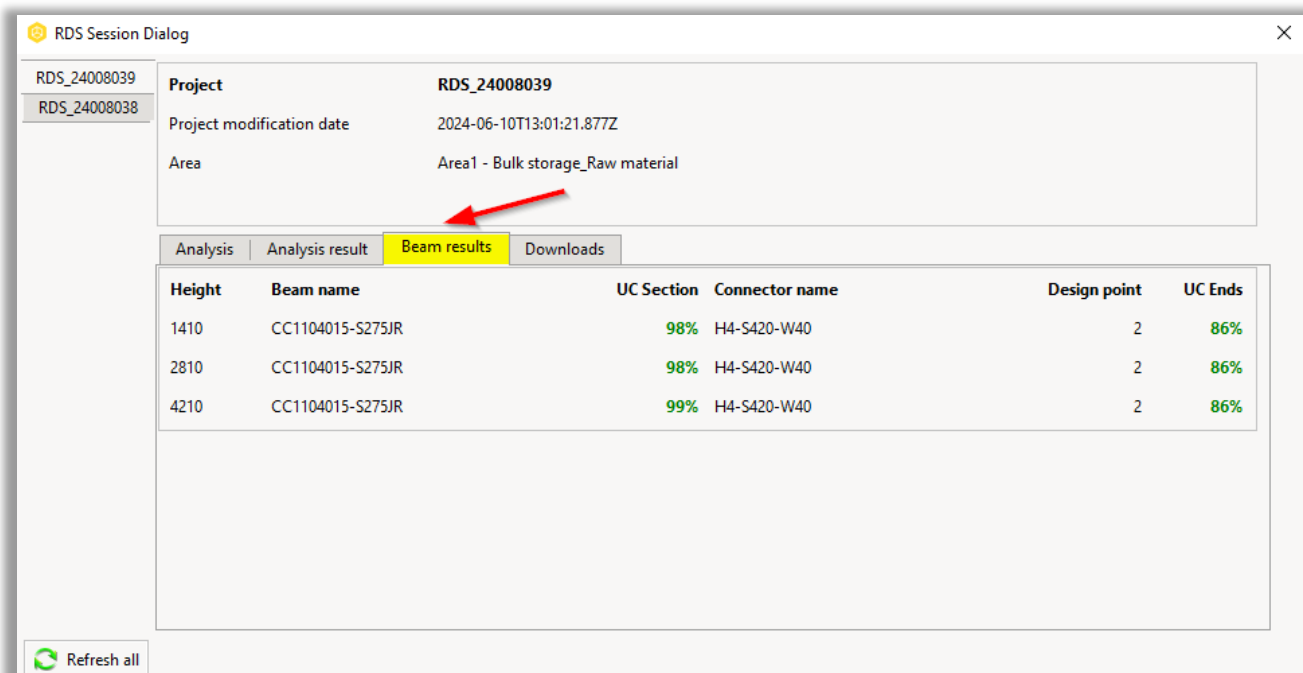
| Cross aisle - Checks on strength |            |        | Down aisle - Checks on strength |            |        |
|----------------------------------|------------|--------|---------------------------------|------------|--------|
| Component                        | UC Section | UC End | Component                       | UC Section | UC End |
| 10068204050PR_M6-S355MC          | 65%        |        | 10068204050PR_M6-S355MC         | 75%        | 14%    |
| U503015_M6-S275JR                | 26%        | 11%    | CC1104015-S275JR                | 99%        | 86%    |

| Cross aisle - Checks on stiffness |     | Down aisle - Checks on stiffness |     |
|-----------------------------------|-----|----------------------------------|-----|
| Deformation                       | UC  | Deformation                      | UC  |
| Upright CA                        | 14% | Upright DA                       | 10% |
|                                   |     | Hook-in beams                    | 99% |

Refresh all

Click tab **Beam result**:



**RDS Session Dialogue**

Project: RDS\_24008039  
Project modification date: 2024-06-10T13:01:21.877Z  
Area: Area1 - Bulk storage\_Raw material

Analysis | Analysis result | **Beam results** | Downloads

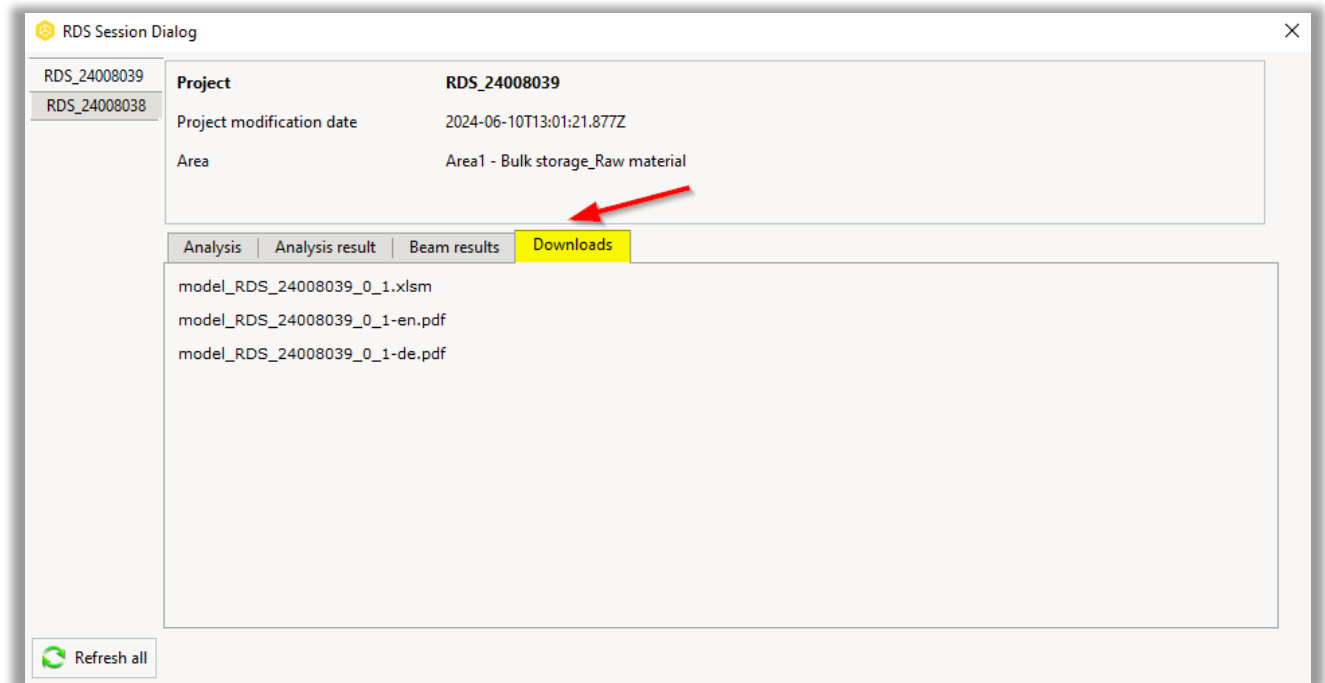
| Height | Beam name        | UC Section | Connector name | Design point | UC Ends |
|--------|------------------|------------|----------------|--------------|---------|
| 1410   | CC1104015-S275JR | 98%        | H4-S420-W40    | 2            | 86%     |
| 2810   | CC1104015-S275JR | 98%        | H4-S420-W40    | 2            | 86%     |
| 4210   | CC1104015-S275JR | 99%        | H4-S420-W40    | 2            | 86%     |

Refresh all

## 2.10 Download analysis results

The RDS Session Dialogue allows also download of analysis results.

Click tab **Downloads**:



All users can download the basic RDS reports in pdf in English and German.

Users with advanced (structural engineering) rights can download also the Posprocessing file in Excel giving verifications of all components.

RDS users with Structural Engineering rights can further download RFEM models in cross-aisle and down-aisle directions directly from RDS (see section 8.3).

## 2.11 Re-opening RDS Session Dialogue

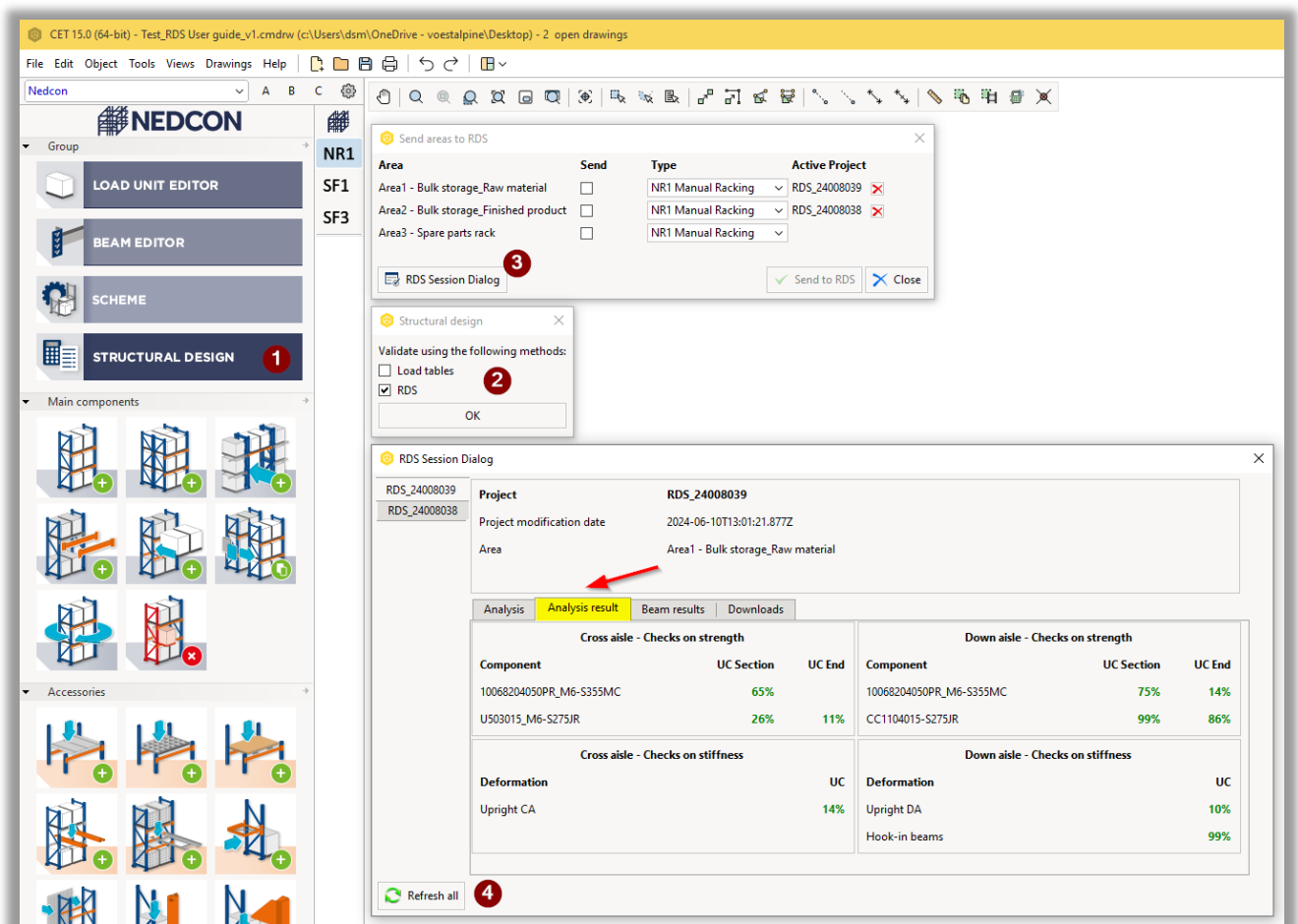
The user may need to re-visit the RDS Session Dialogue either during a CSA session, where the analysis has been performed, or during a later (new) session.

1. Click on **STRUCTURAL DESIGN** (1),
2. Select **RDS** analysis (2) and confirm **OK**,
3. Click **RDS Session Dialogue** (3).

In case of CSA session, where the analysis has been performed, the results are present in CSA and they are shown directly.

In case of a later (new) session, CSA knows only the RDS project number, but the results are stored in RDS. They need to be fetched first.

4. Click **Refresh all** (4). The results then, after signing in RDS, appear.



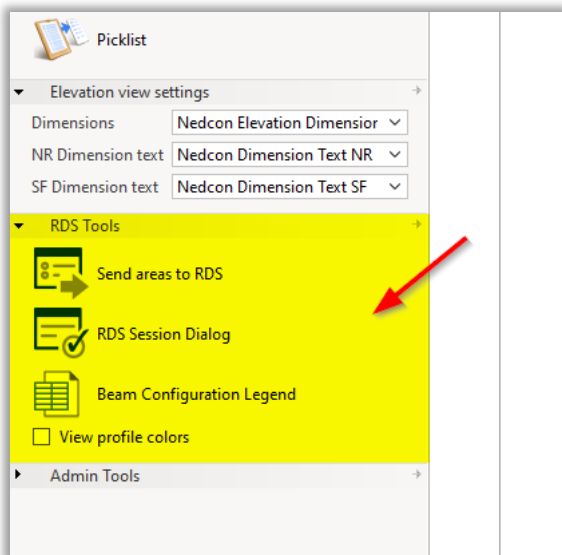
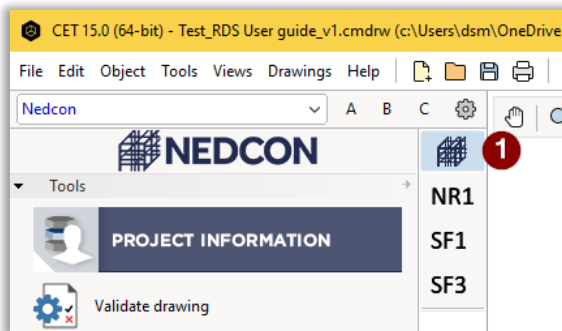
## 2.12 Secondary location of Advanced mode control elements

All controls necessary to control structural engineering design in Advanced mode are repeated once again in the tab Nedcon.

The reason is to allow access to these controls also from other tabs than just from the tab NR1.

The controls can be accessed as follows:

1. Click tab **Nedcon** (1),
2. Select any of the control elements in the section **RDS Tools**.



## 3 Beam Editor

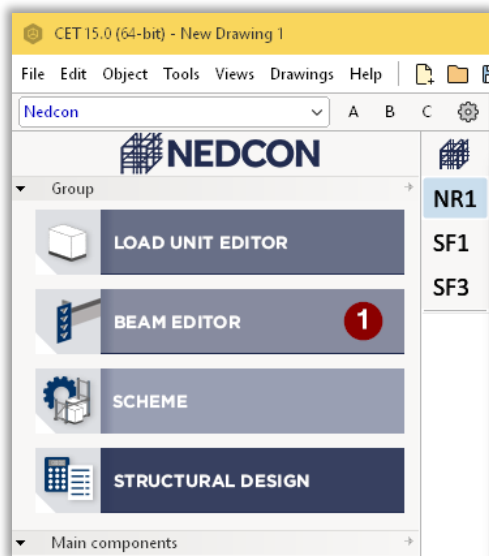
### 3.1 General

The purpose of the Beam Editor is to define beam profiles to be used in RDS verifications.

The defined profiles can be then assigned to individual beams in the structure. See the next sections on how to do that. The profiles cannot be assigned from the Beam Editor.

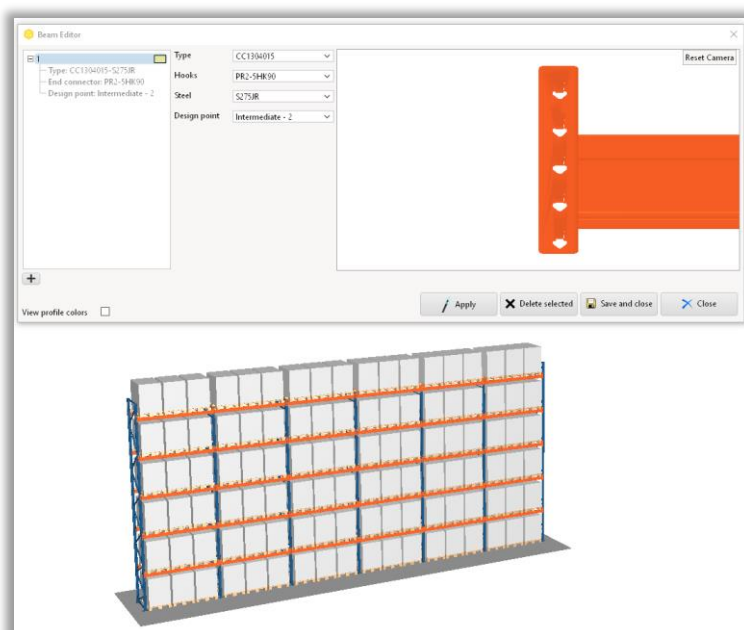
#### Opening the Beam Editor

1. Make sure you are operating in the **Advanced design mode**,
2. Click the **Beam Editor** button (1).



### 3.2 Initial beam profiles

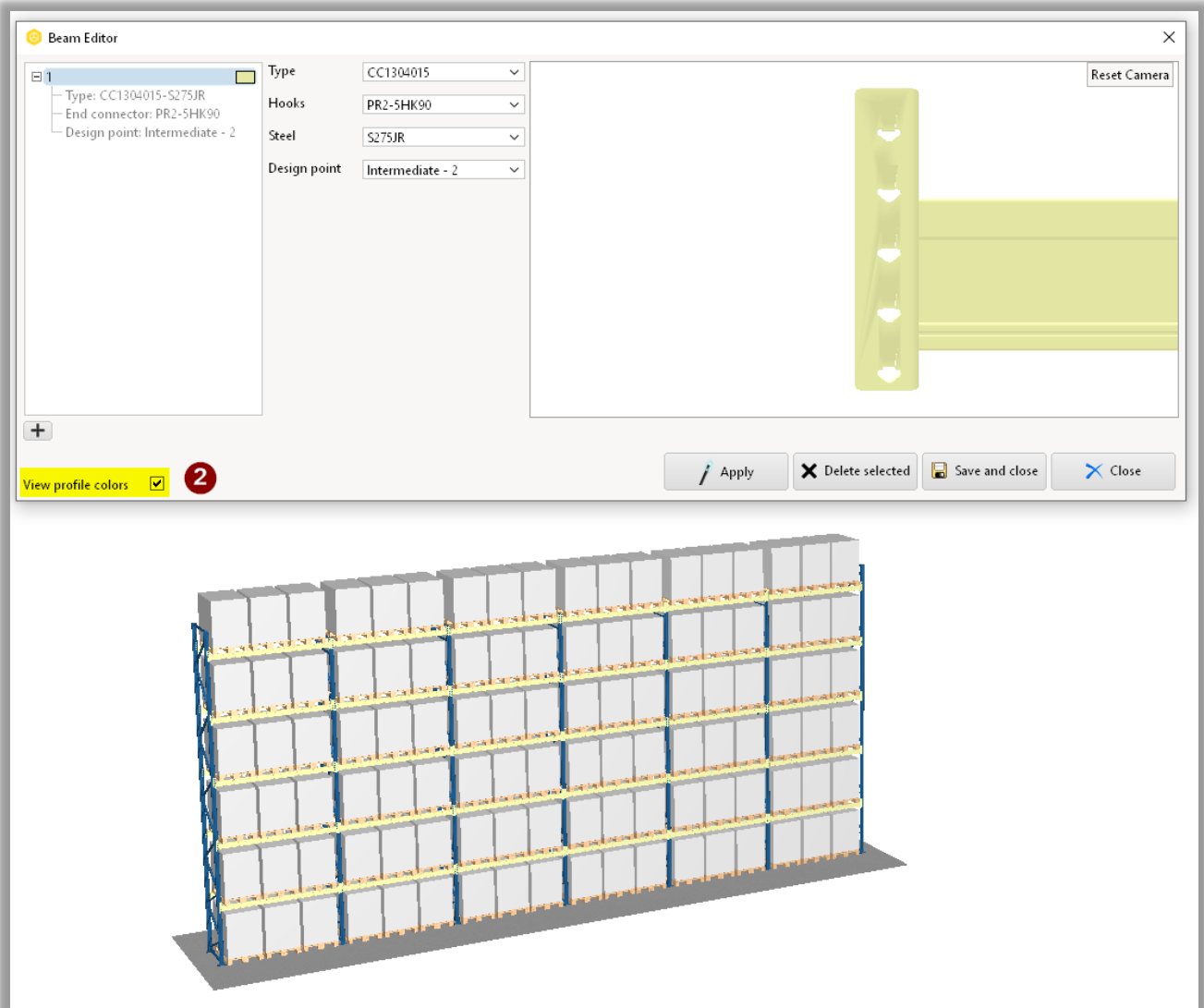
On opening, the beam editor shows all beam profile currently present in the model:



### 3.3 Displaying beam profiles in 3D model

In order to find out what beam profiles are where, the user can switch on displaying of colour coding of beam profiles in the model:

1. Click **View profile colors** (2).

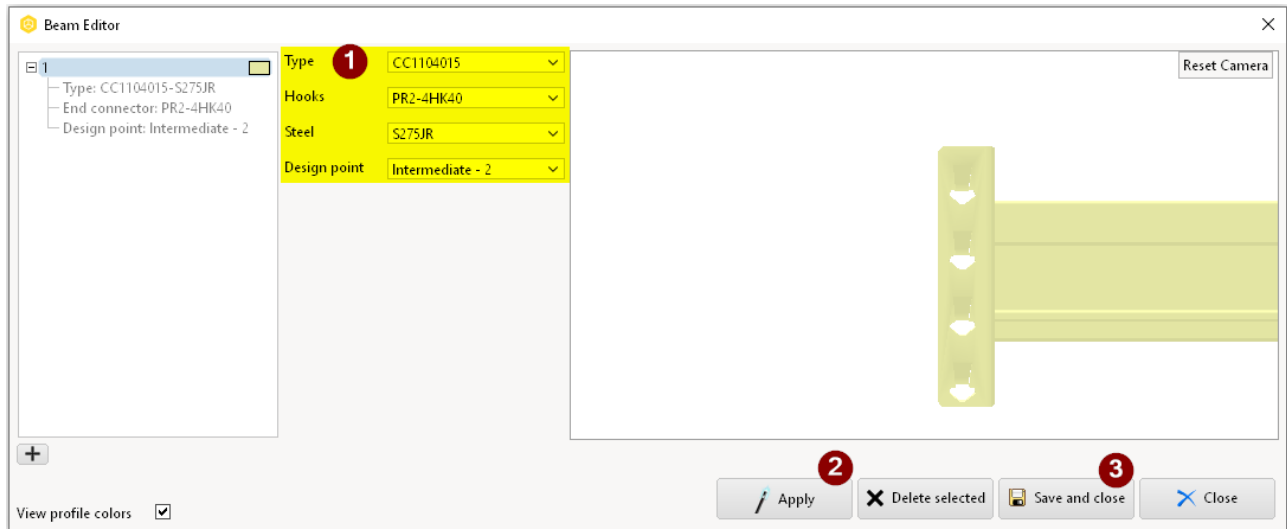


The manufacturing colour of the real beam stays unchanged (as listed in BOM).

### 3.4 Modifying existing beam profiles

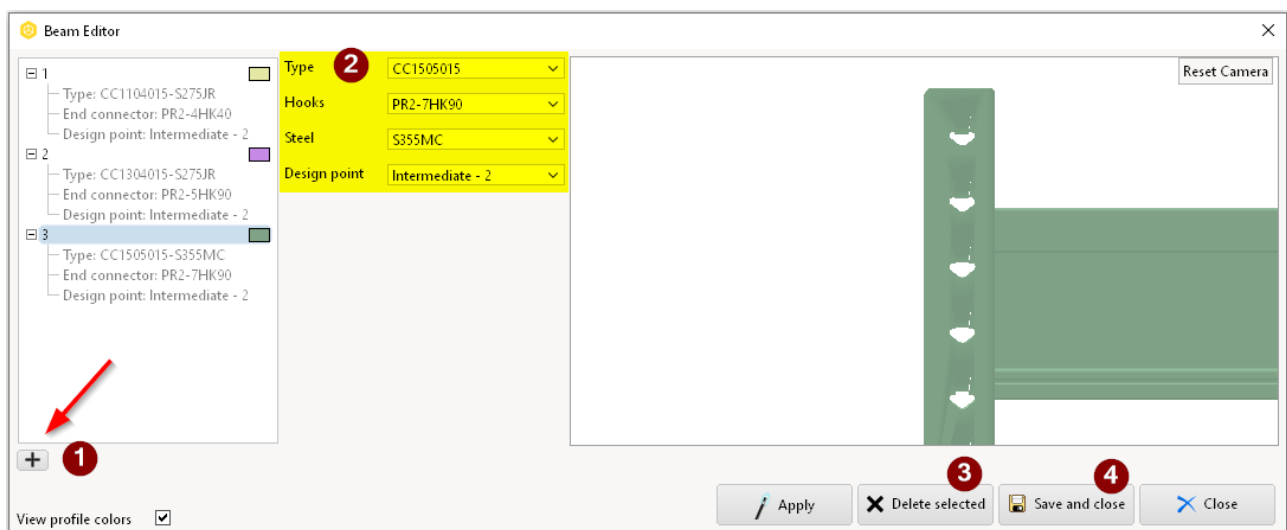
1. Modify the selected **beam's configuration** (1) using the pull-down menus,
2. Click **Apply** (2), if you wish to apply the changes already when in the Beam Editor,
3. Click **Save and close** (3).

Note, that step (2) can be omitted. The profiles will be then applied when closing the Beam Editor.



### 3.5 Define new beam profiles (custom beam profiles)

1. Click the **plus sign** (1) to add a new beam profile,
2. Modify the selected **beam's configuration** (2) using the pull-down menus,
3. Multiple new beams can be added at the same time,
4. Selected beam profiles can be removed by clicking **Delete selected** (3),
5. Click **Save and close** (4).

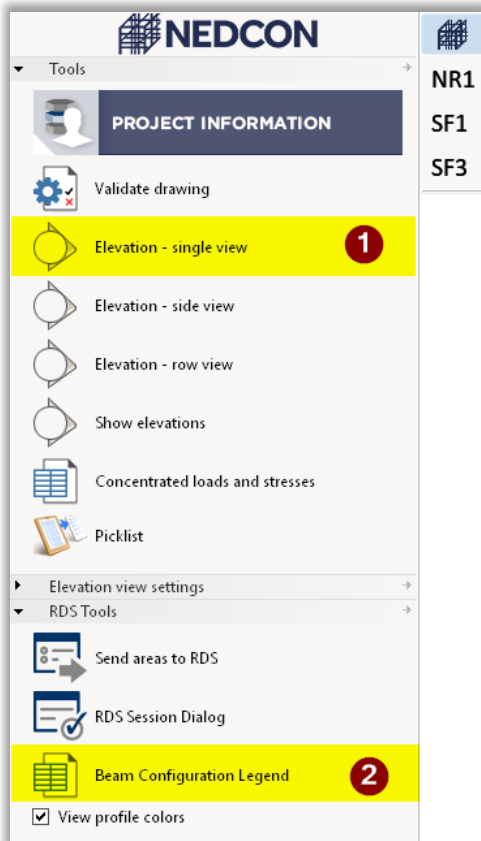


Note, that only beam profiles which are not used in the drawing can be deleted.

### 3.6 Displaying beam profiles in 2D view

Similarly to 3D, in order to find out what beam profiles are where in 2D, the user can use beam profile numbers in Elevation – Single views and the associated Beam Configuration Legend:

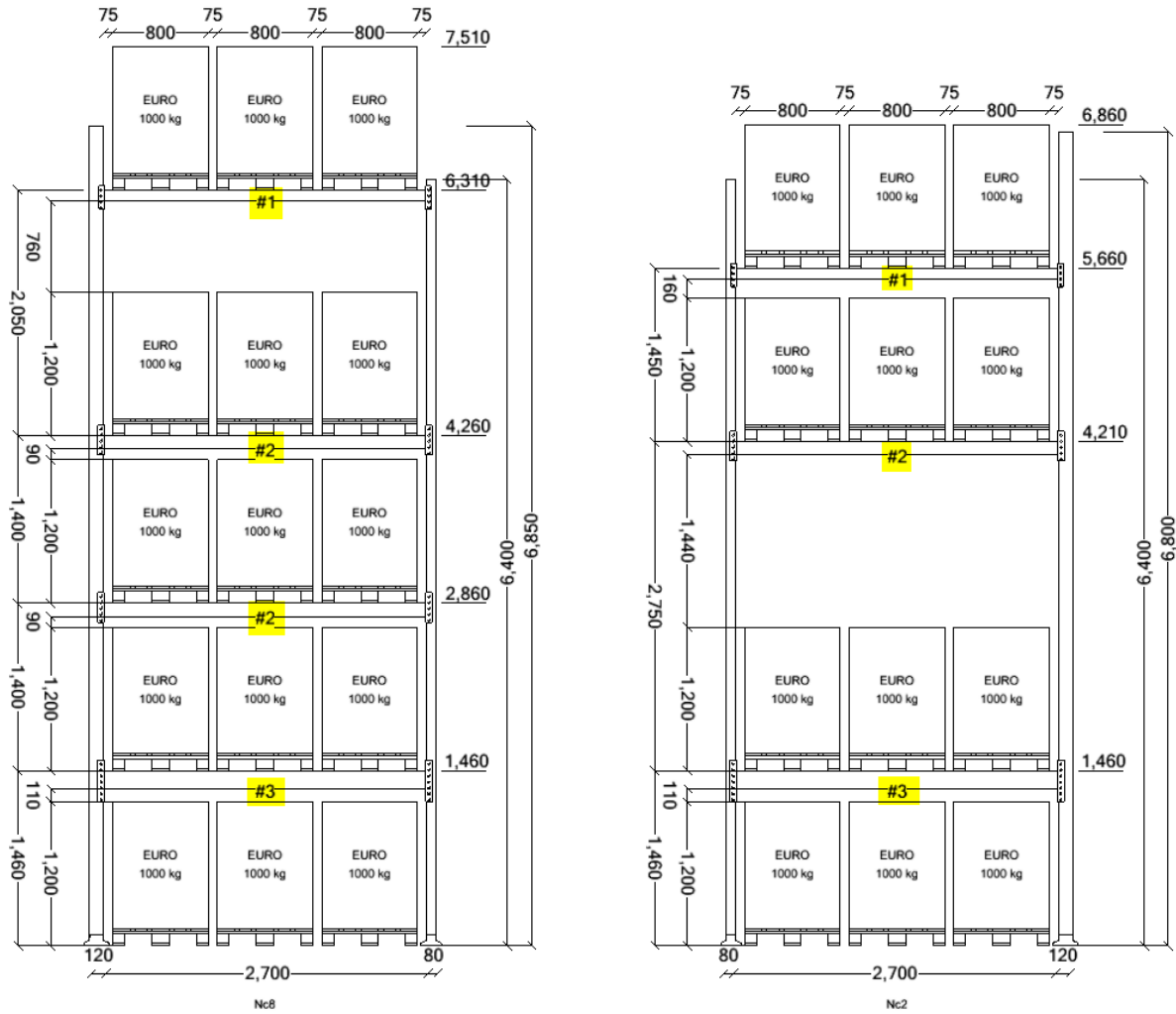
1. Insert one or multiple **Elevation – Single views** (1) into 2D in the usual way,
2. Click **Beam Configuraton Legend** (2) and add the table to 2D.



The Single views feature beam profile numbers superimposed over beams.

The Legend gives description of each beam profile. It can be scaled using its Quick Property Box.

Example:

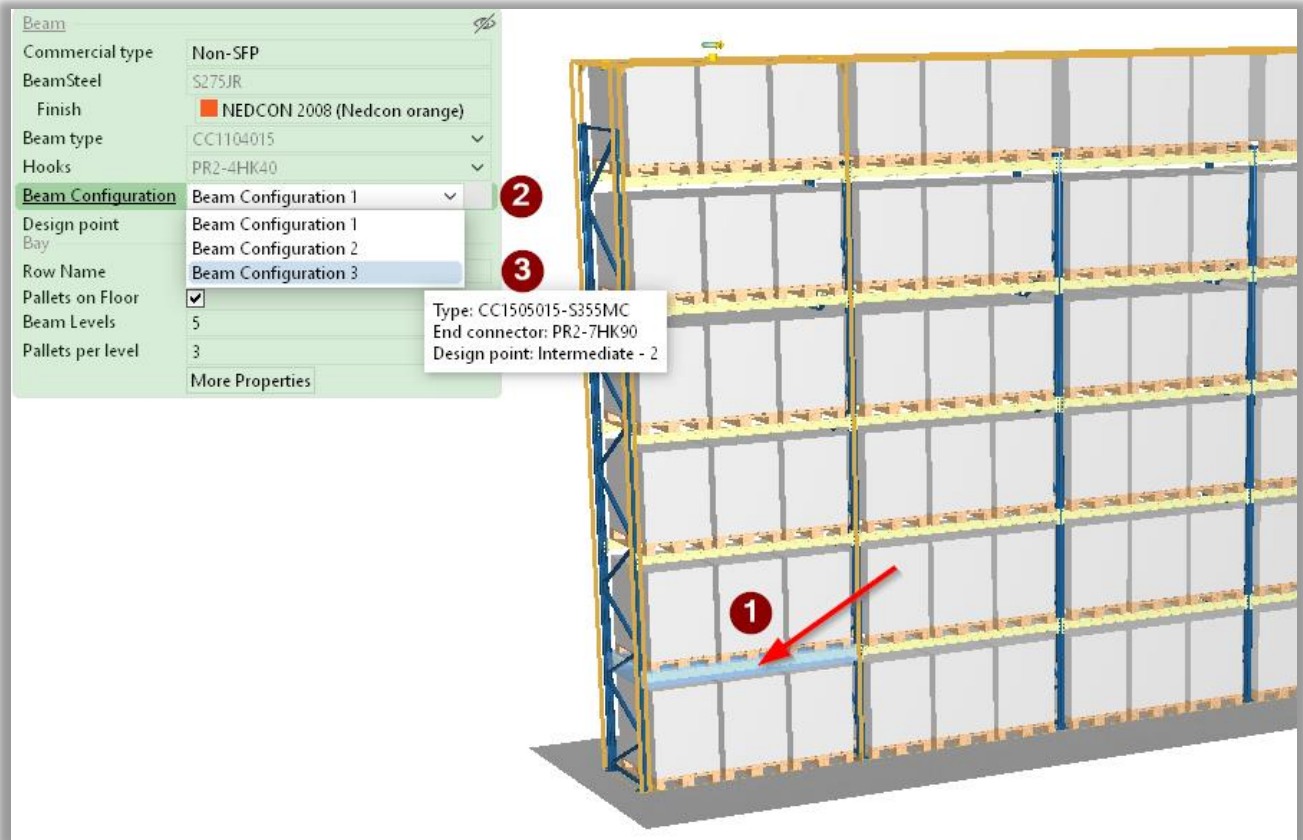


| Beam Configurations |                  |           |                  |
|---------------------|------------------|-----------|------------------|
| 1                   | CC0904015-S275JR | PR2-4HK40 | Intermediate - 2 |
| 2                   | CC1104015-S275JR | PR2-5HK90 | Intermediate - 2 |
| 3                   | CC1505015-S355MC | PR2-7HK90 | Intermediate - 2 |

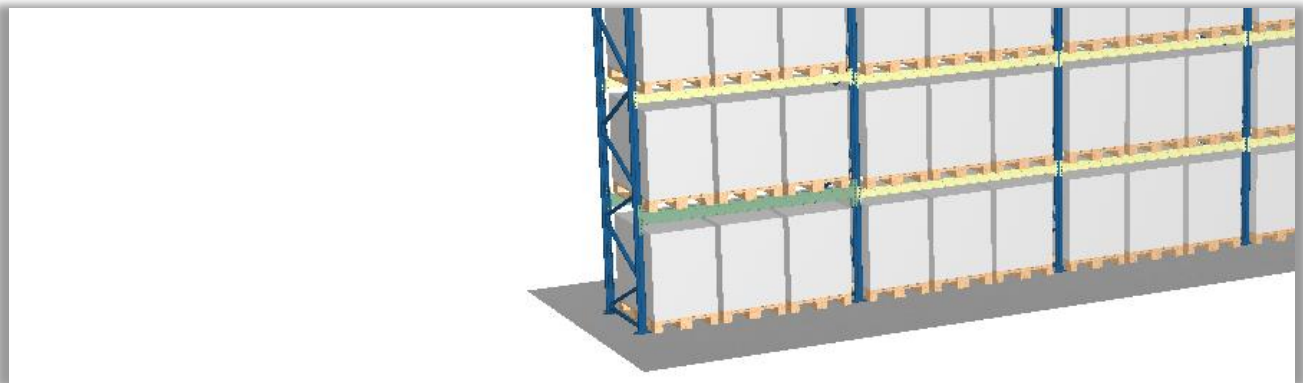
## 4 Assigning beam profiles

### 4.1 Using Quick property box and Copy configuration tool

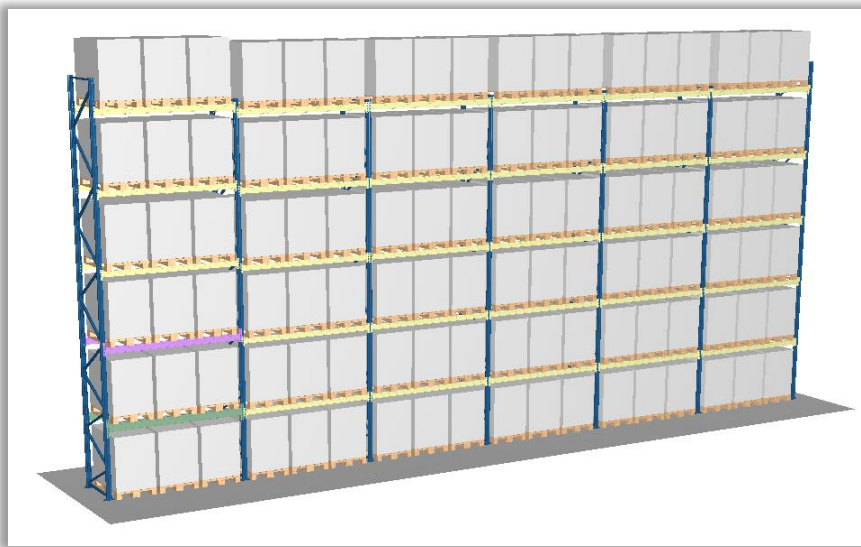
1. Select a pair of beam by clicking **directly on the beam** (1),
2. The green Quick property box opens,
3. Click on **Beam Configuration** pull-down menu (2),
4. When hovering above beam profiles, a pop-up giving the configuration details appears,
5. Select the desired **beam profile** (3),
6. The beam profile is directly assigned to the selected pair of beams.



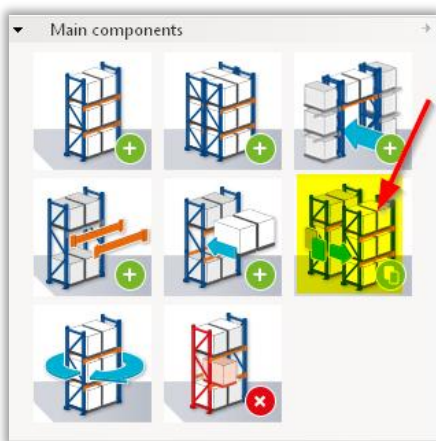
Resulting structure:



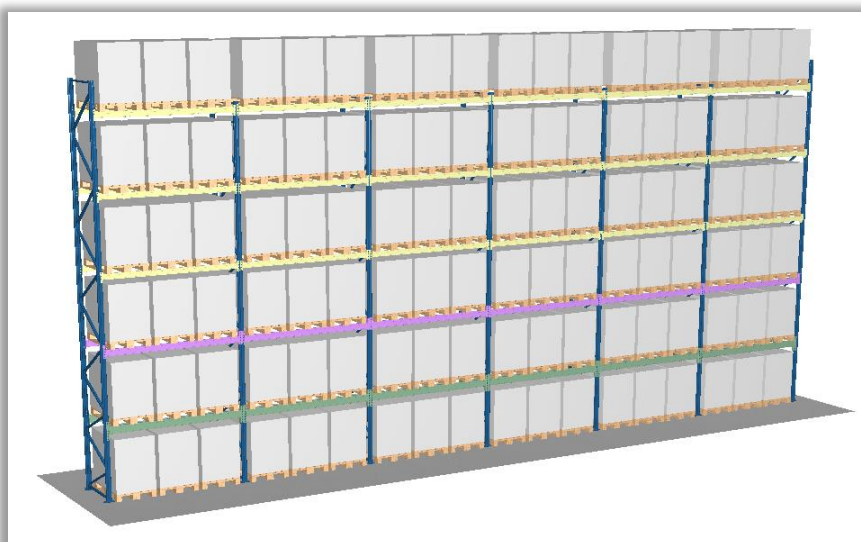
Repeat the process for all beams in the initial (source) bay, where applicable:



Use the **Copy configuration tool** to spread the bay configuration including the beam profiles to the rest of the structure, where applicable:



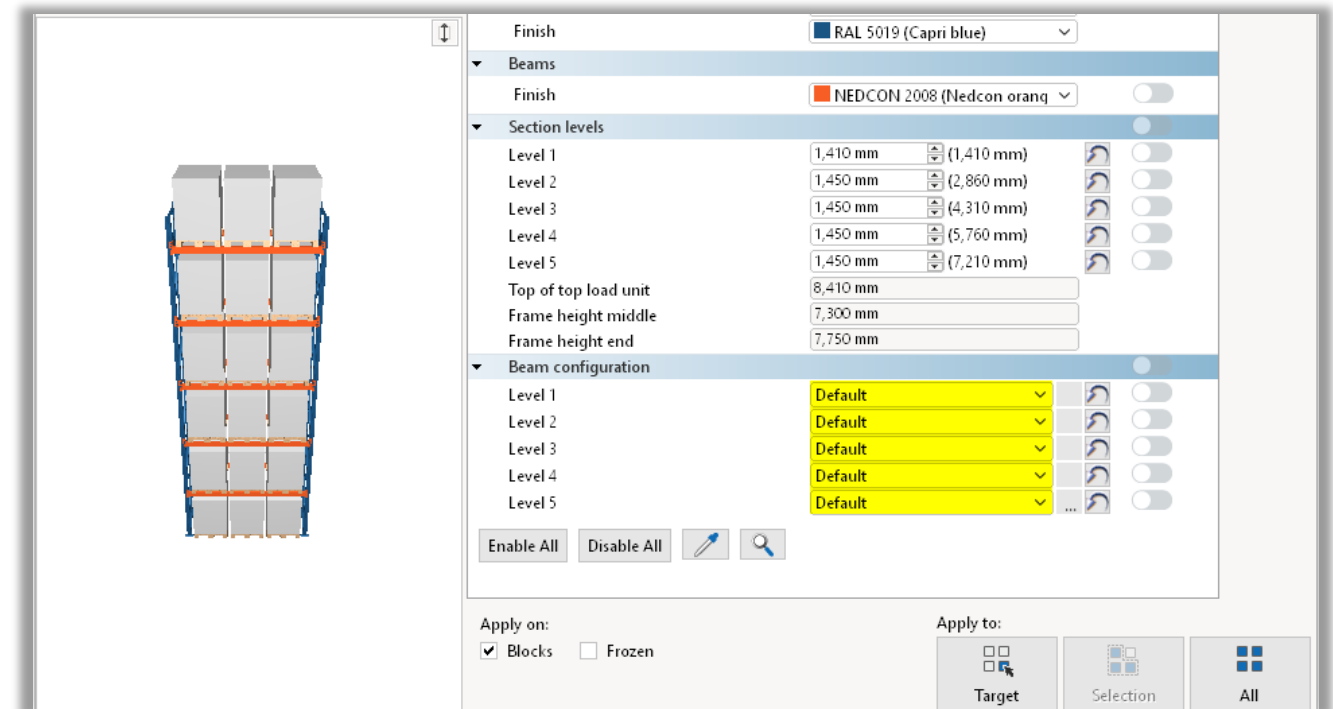
Resulting structure:



## 4.2 Using Scheme

### a1. Default beam profiles

When initially opening the Scheme, the section **Beam configuration** shows Default beam profiles on all levels:



The Default beam profile is a beam profile estimated by the application based on the choices made in the Scheme.

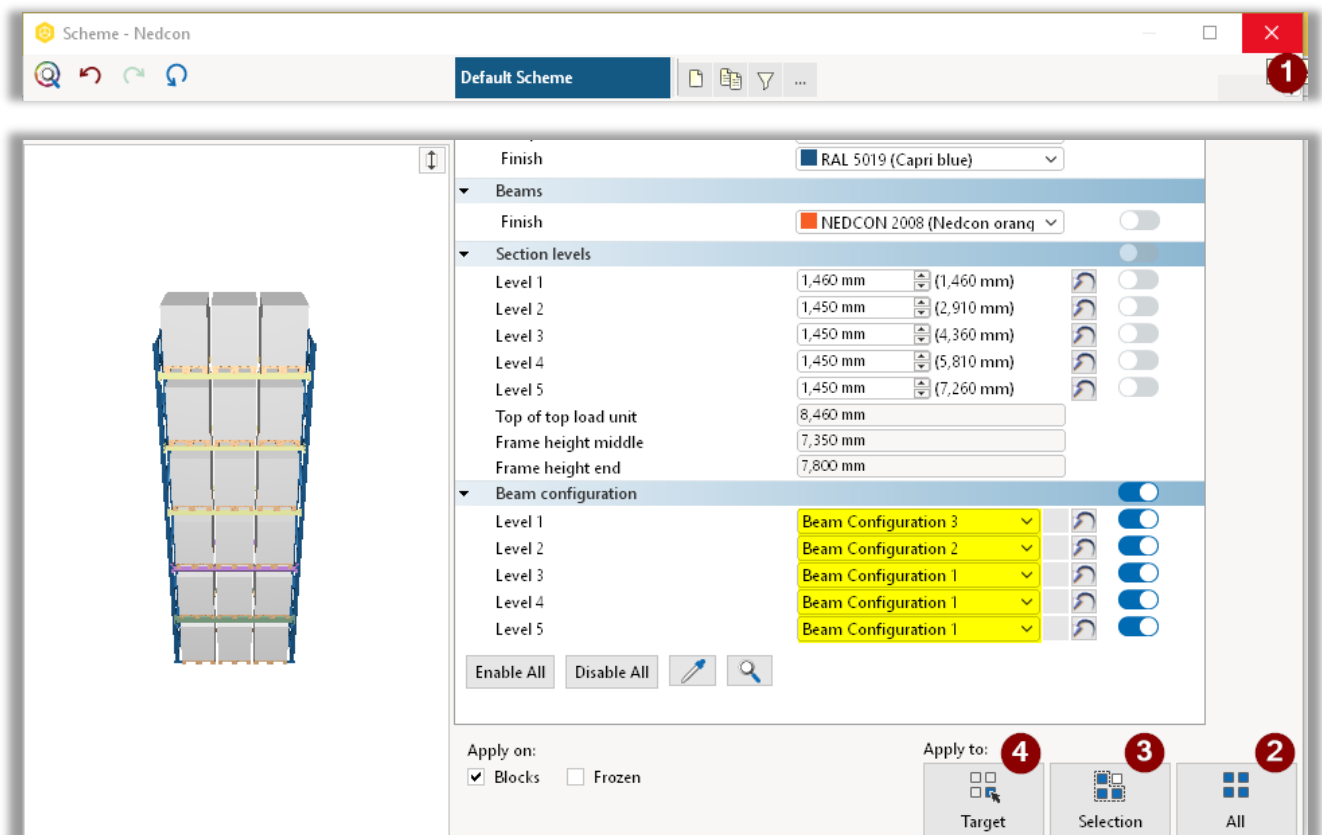
For example, the application estimates lower beam profiles for rack configurations with less pallets per compartment and less levels and higher beam profiles for rack configurations with more pallets per compartment and more beam levels.

The Default beam profiles however stay just an estimate. **Every design needs to be verified and optimised in every case using the STRUCTURAL DESIGN function.**



## a2. Beam profiles assigned by user

The user can assign own beam profiles, which have been defined previously in the Beam Editor, to individual beam levels using the pull-down menus in section **Beam configuration**:



## b. Applying beam profiles – 4 different ways

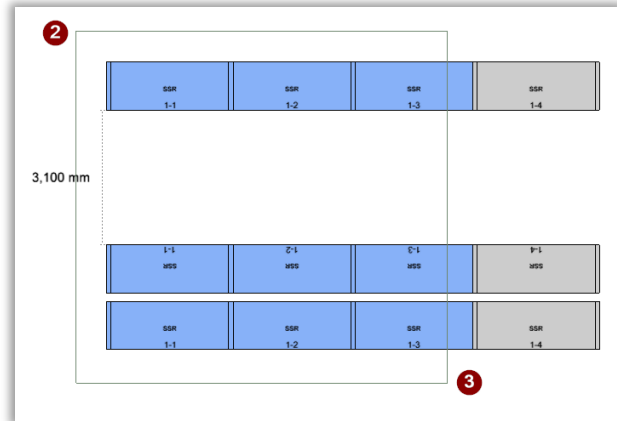
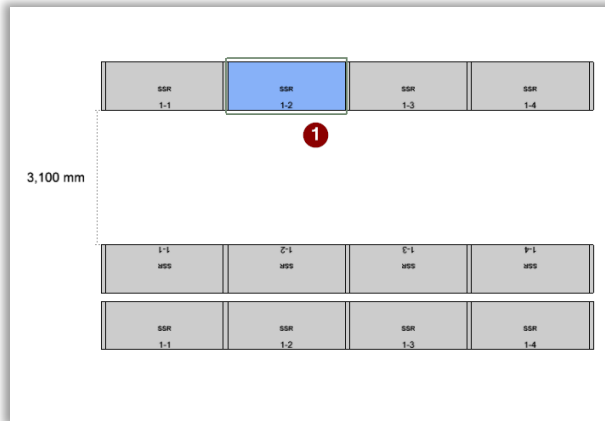
The beam profiles can be then applied in 4 different ways:

- **Closing the Scheme (1)** – clicking the red cross in upper right-hand corner,
- The beam profiles are applied only on newly created rows, i.e. rows inserted after closing the Scheme,
- **Applying to All (2)**
- The beam profiles are applied on all existing rows and on all newly created rows after closing Scheme,
- **Applying to Selection (3)**
- The beam profiles are applied only on already selected rows or their selected portions,
- **Applying to Target (4)**
- The beam profiles are applied only on rows or their portions designated by the target tool.

### c. Target tool

The target tool can be used as:

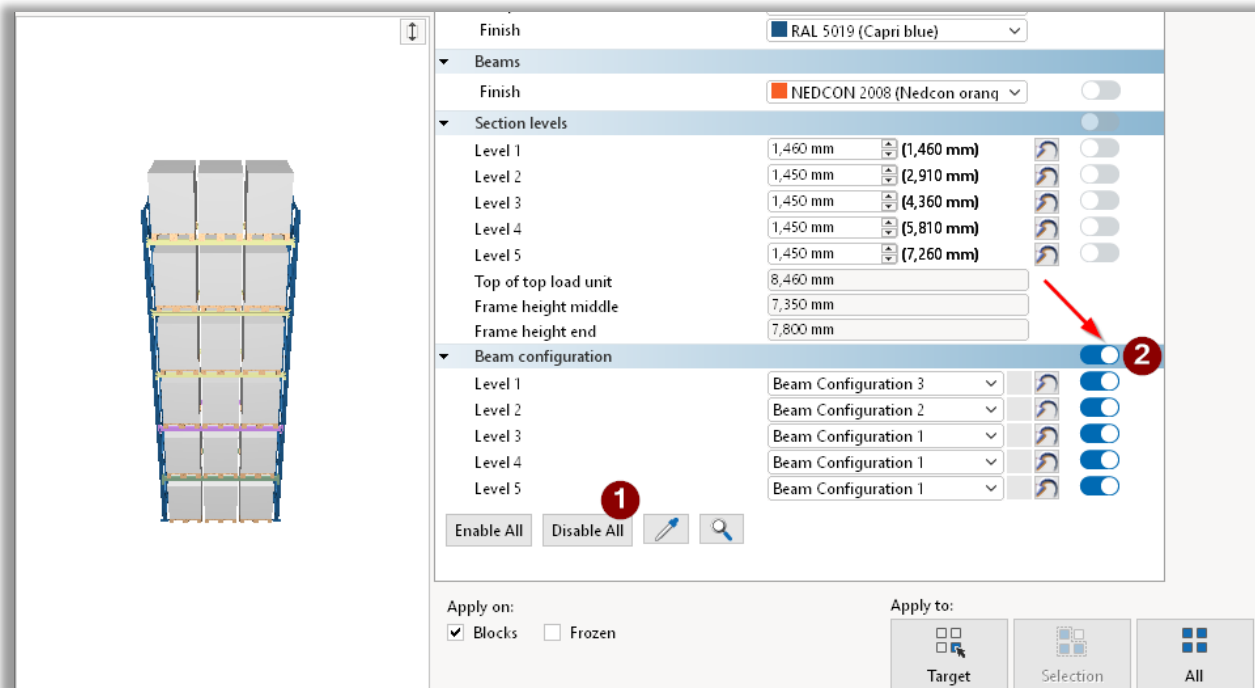
- single bay selector - click once on individual bays (1) ,
- selection rectangle - select a row or its portion by clicking the positions of opposite corners of the selection rectangle (2, 3).



### d. Applying beam profiles only

Note that if the user wishes to apply only the beam profiles and not the other properties present in the Scheme, he or she then needs to:

1. Click on **Disable All** (1) to switch off application of all properties in the Scheme,
2. Click on the **Beam configuration switch** (2) to switch on application of beam profiles only.

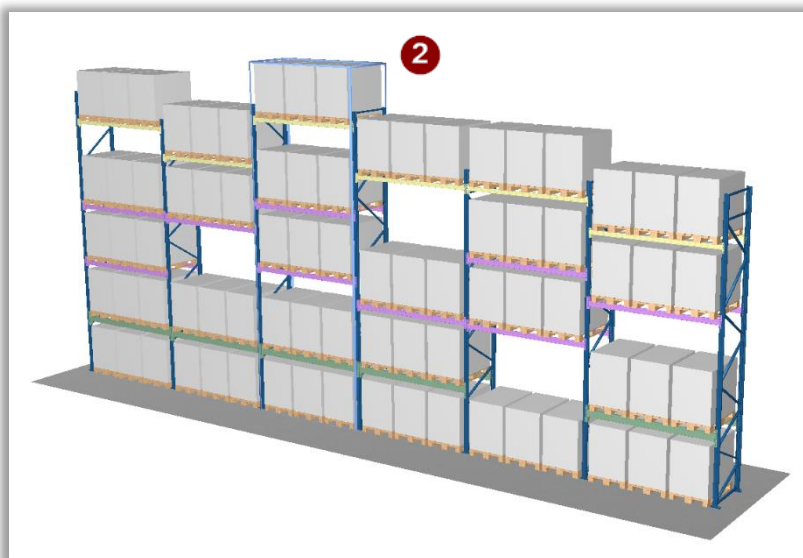
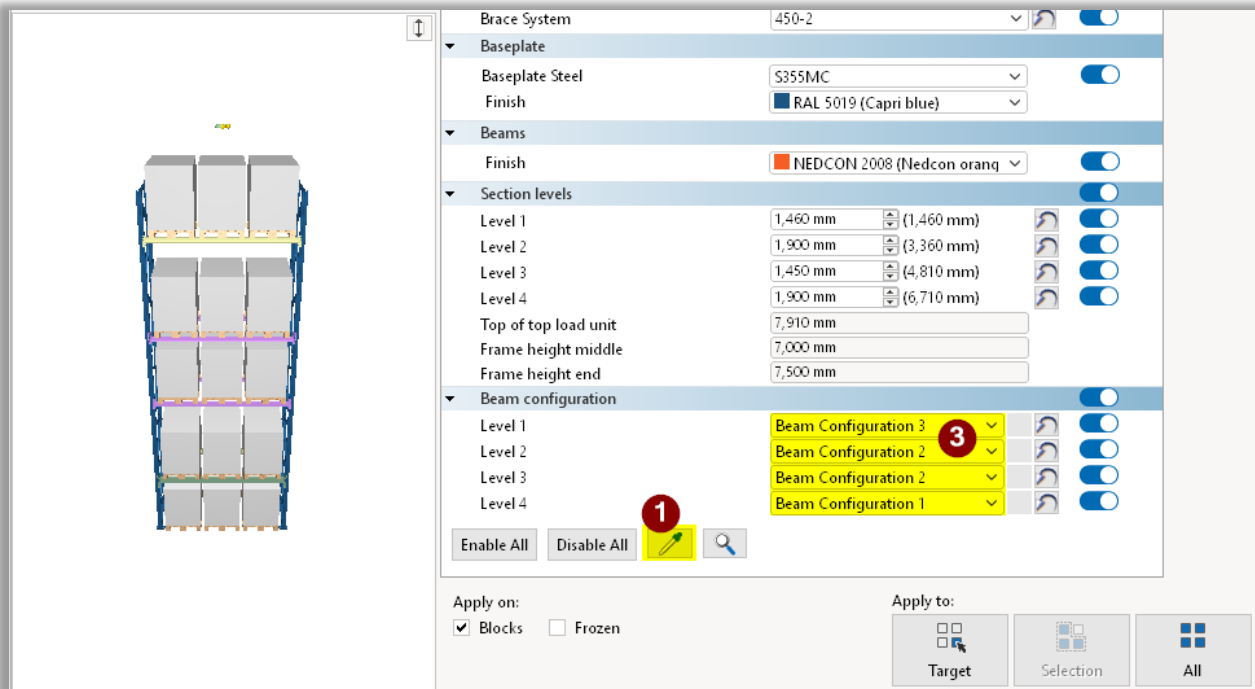


### e. Eye-dropper tool

Finally, when using the Scheme to assign beam profiles, it is handy to be able to load the already existing bay configuration incl. beam profiles in order to be able to modify it.

This can be achieved by using the Eye-dropper tool:

1. Open the **Scheme** and click the **Eye-dropper tool** (1),
2. Use your mouse to control the **light-blue bay selector** (2) to select the source bay configuration,
3. Go to section **Beam configuration** and modify **beam profiles per level** (3),
4. Apply the beam profiles using one of the **4 different ways** described above.



## 5 Assigning upright frame profiles

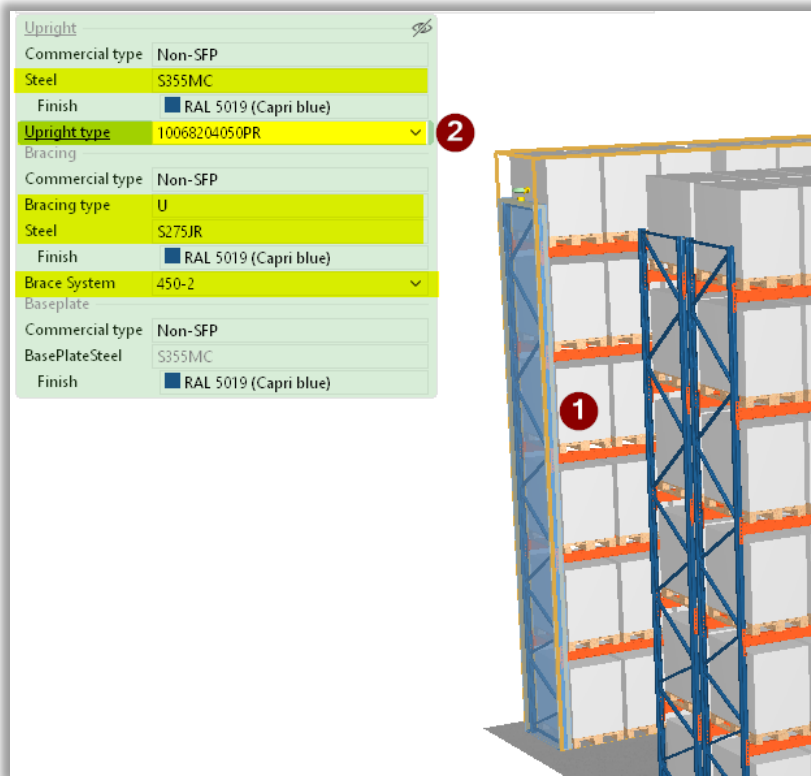
### 5.1 Using Quick property box

The upright frame profiles configuration consists of, where applicable:

- Upright profile, (from 08067204050PR to 12083255070PR),
- Upright steel grade, (S355MC for S350GD),
- Frame bracing profile (U or OB),
- Frame bracing steel grade (S275JR or S250GDS),
- Frame arrangement (450-2 or 600-2).

Any element of the configuration can be changed using the Quick property box:

1. Select an upright frame, e.g. by clicking **directly on the frame** (1),
2. The green Quick property box opens,
3. Change any of upright frame profiles **configuration elements** (2).
4. The changes are directly applied to the selected upright frame.



#### Note

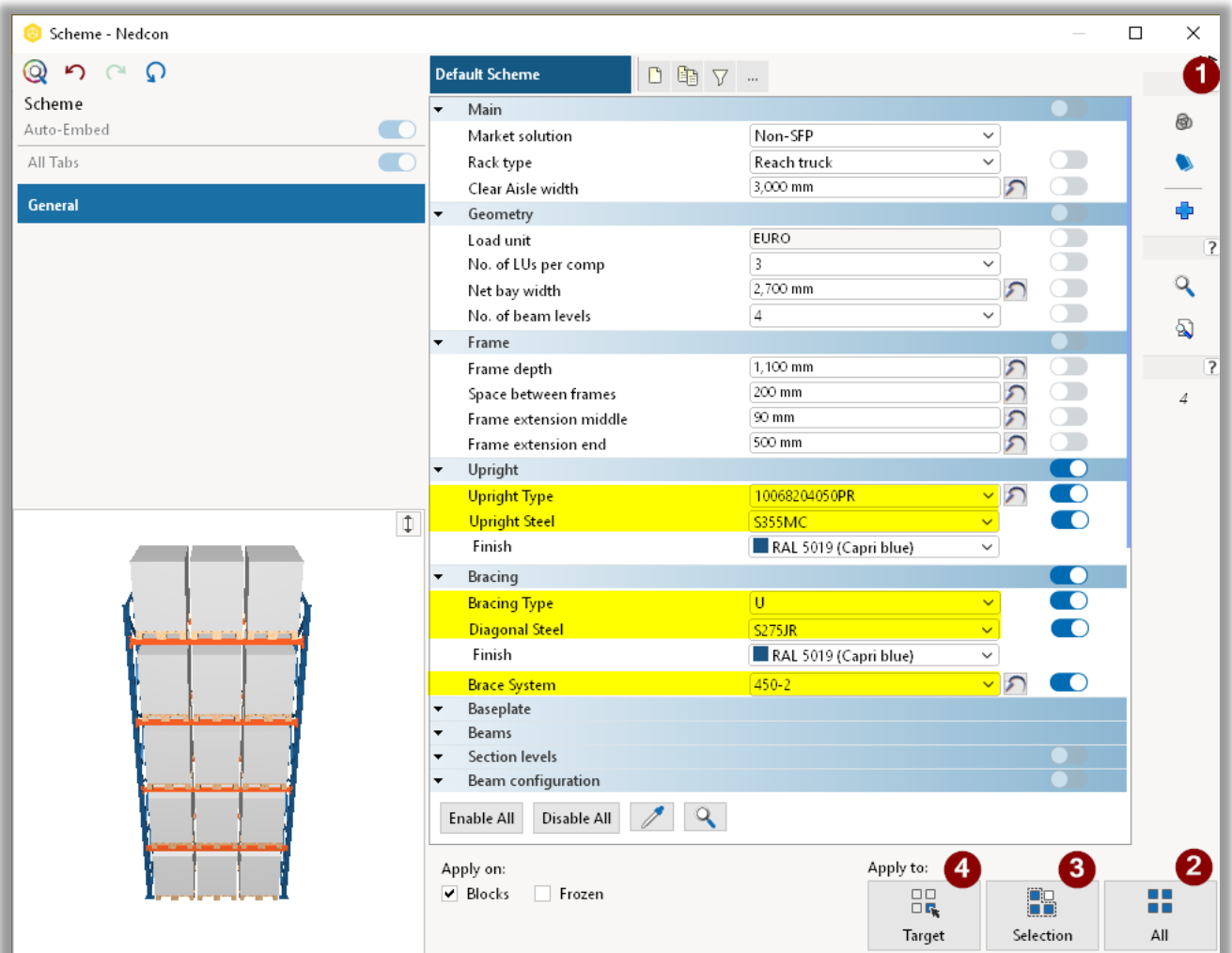
The Quick property box is suitable for changing profiles of one or a few upright frames only. It is not suitable for changing profiles of many upright frames.

Use the Scheme instead to assign upright frame profiles to larger volume of upright frames.

The Copy configuration tool can't be used to copy upright frame profiles. It copies bay configuration and beam profiles only.

## 5.2 Using Scheme

The user can assign own upright frame profiles to upright frames using the pull-down menus in sections **Upright** and **Bracing**:



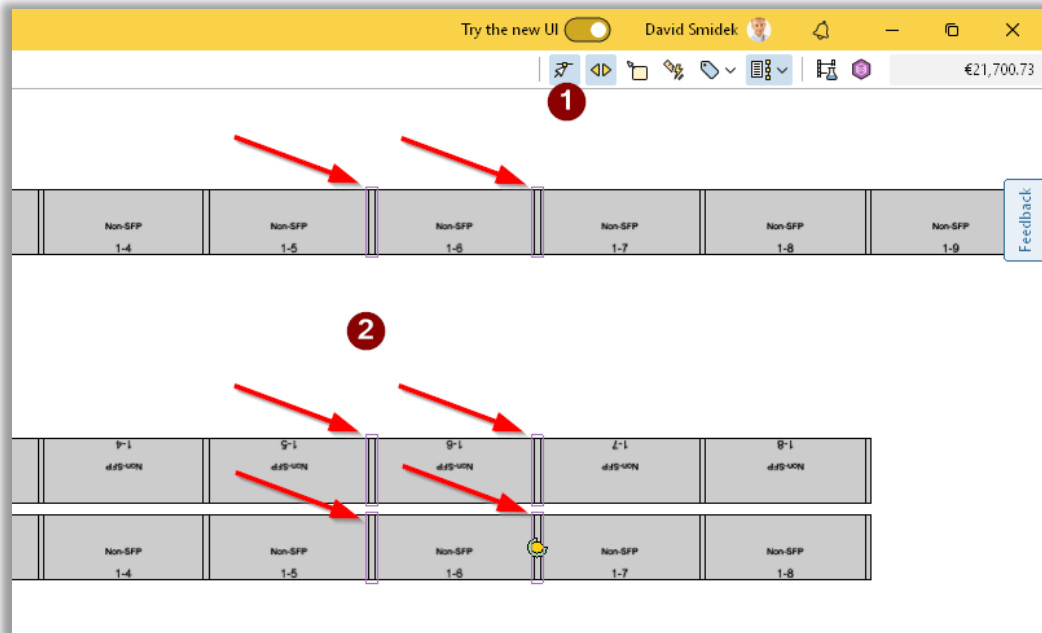
The upright frame profiles can be then applied in 4 different ways:

- **Closing the Scheme (1)** – clicking the red cross in upper right-hand corner,
- The upright frame profiles are applied only on newly created rows, i.e. rows inserted after closing the Scheme,
- **Applying to All (2)**
- The upright frame profiles are applied on all existing rows and on all newly created rows after closing the Scheme,
- **Applying to Selection (3)**
- The upright frame profiles are applied only on already selected rows or their selected portions (i.e. on selected upright frames). See Guide 1.
- **Applying to Target (4)**
- The upright frame profiles are applied only on rows or their portions (i.e. on selected upright frames) designated by the target tool. See Guide 2.

## Guide 1 – Selecting multiple upright frames

To select multiple upright frames, do the following:

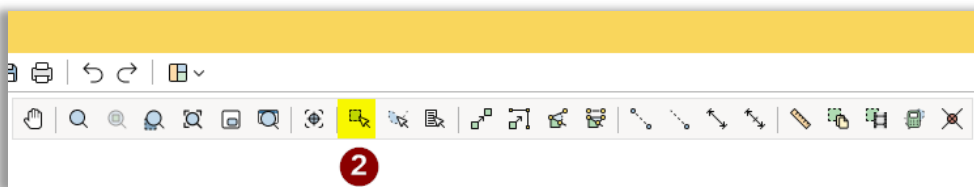
1. Switch to **single select mode** (1) – the switch turns blue,
2. Hold **Ctrl** pressed and select the frames by **clicking** on them (2).
3. The already selected upright frames get a **purple rectangle** directly around them (see the red arrows).

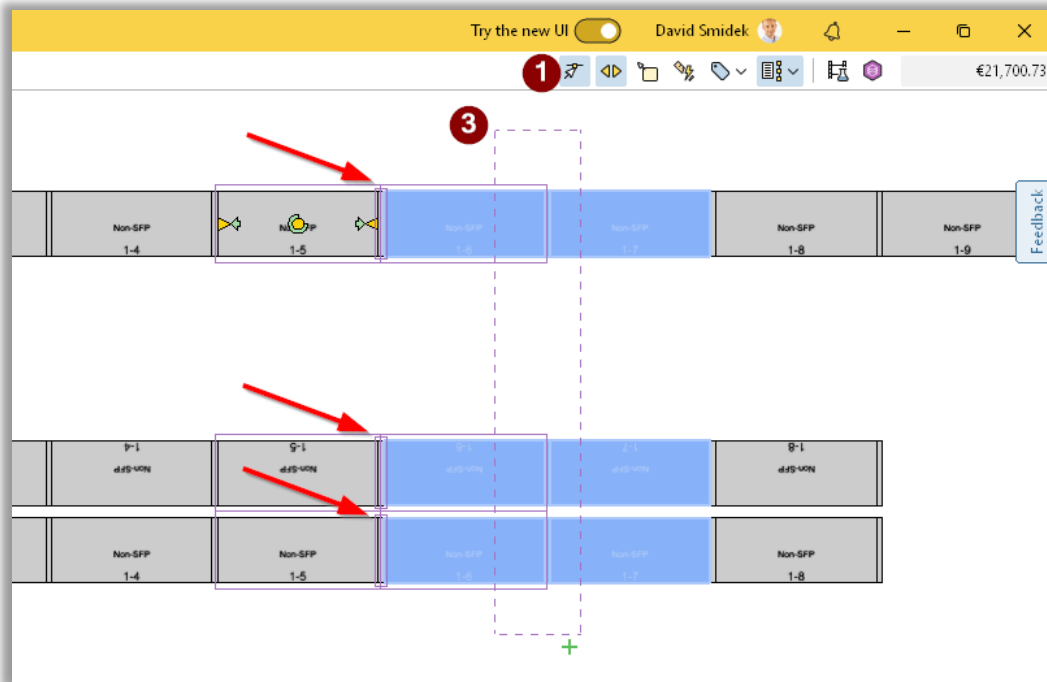


Alternatively, you can use a selection rectangle:

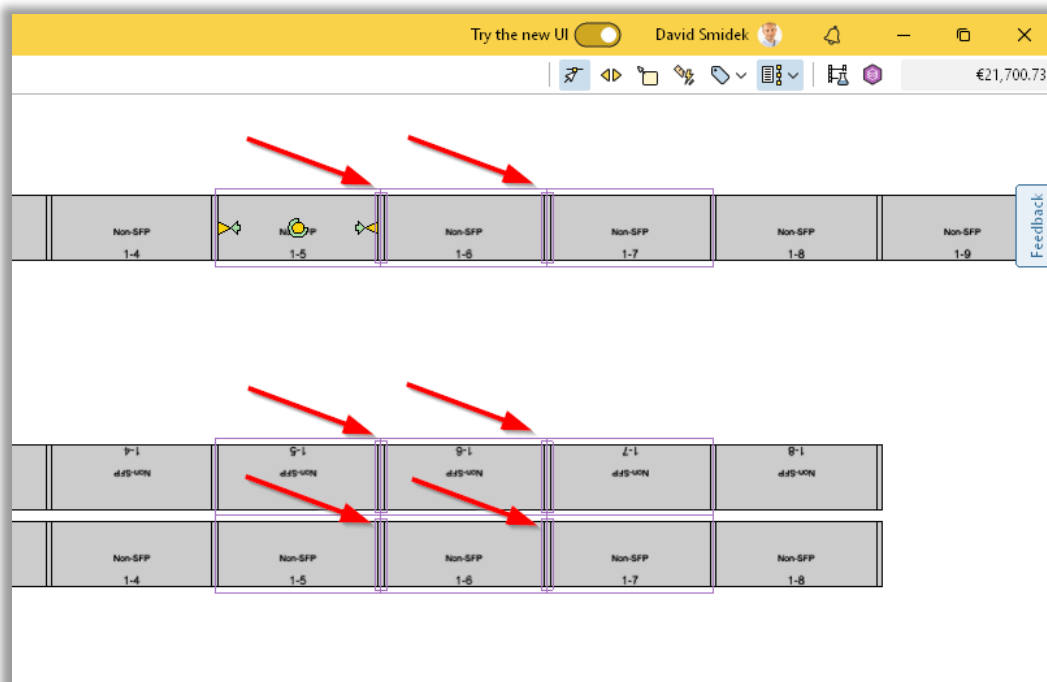
1. Switch to **single select mode** (1) – the switch turns blue,
2. Click on **Rectangle select** (2) – the cursor gets a purple selection rectangle,
3. Select multiple upright frames by placing a **selection rectangle around them** (3),
4. The already selected upright frames get a **purple rectangle** directly around them (see the red arrows).

**Note** - You may neglect the purple rectangles around the bays. They are generated automatically, but they are not relevant for assigning profiles to upright frames.





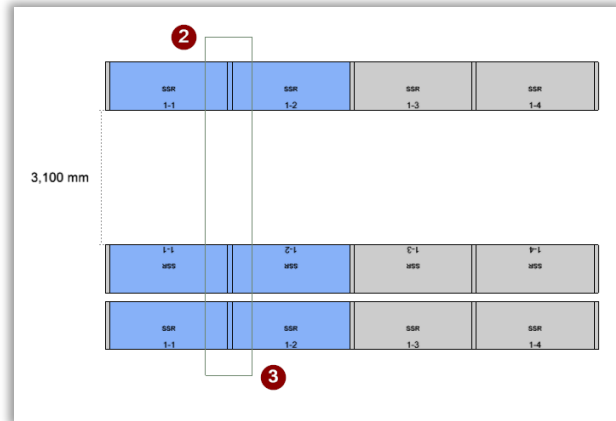
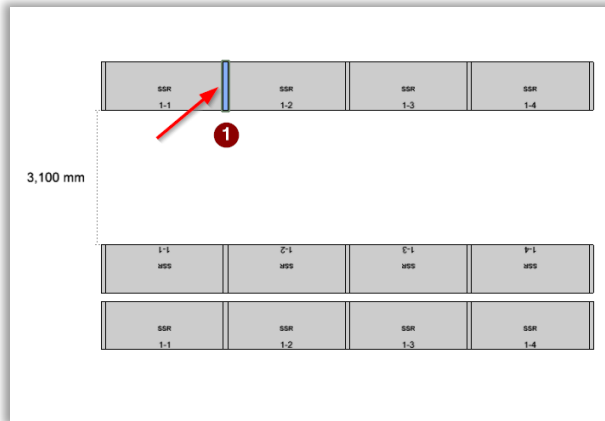
Resulting selected frames (6 frames have been selected):



## Guide 2 – Using the target tool

The target tool can be used as:

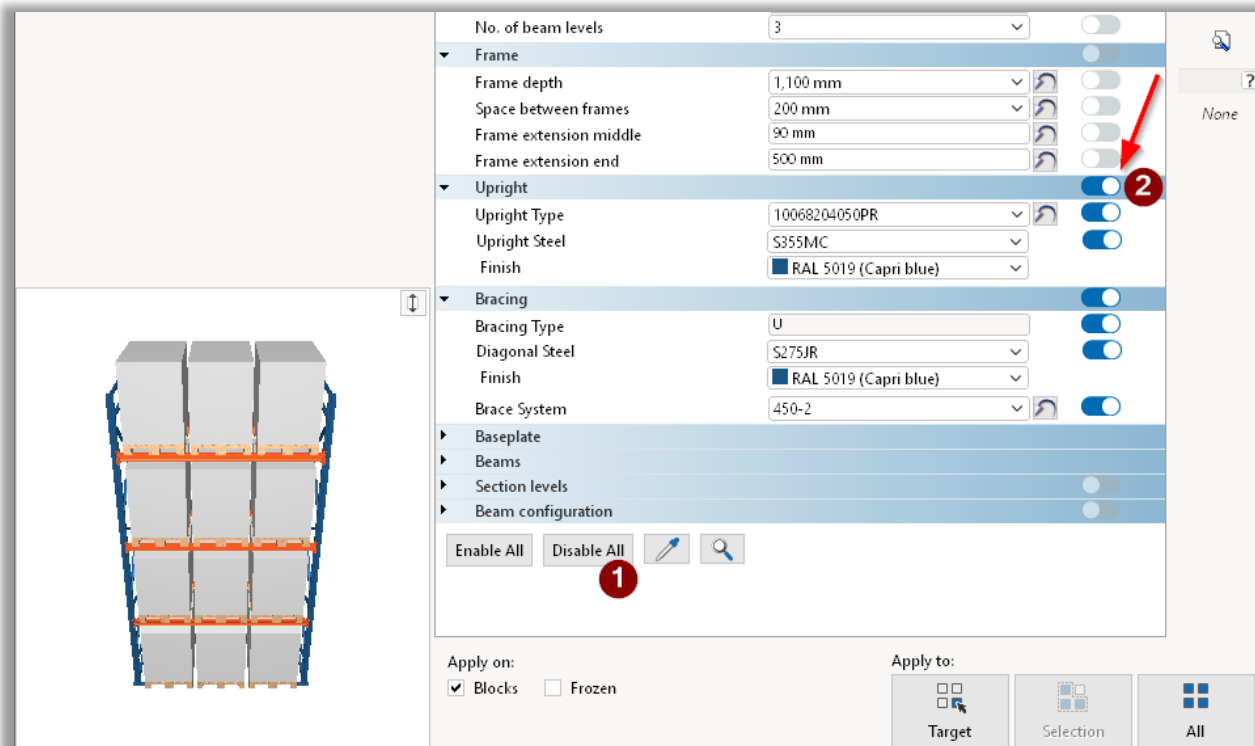
- single upright frame selector - click once on individual upright frames (1),
- selection rectangle - select multiple upright frames by clicking the positions of opposite corners of the selection rectangle (2, 3).



## Applying upright frame profiles only

Note that if the user wishes to apply only the upright frame profiles and not the other properties present in the Scheme, he or she then needs to:

1. Click on **Disable All** (1) to switch off application of all properties in the Scheme,
2. Click on the **Upright** and **Bracing** switches (2) to switch on application of upright frame profiles only.



## 6 Further operations with projects

### 6.1 Changing structure with existing RDS results

If the user changes any of the elements defining a structure for RDS analysis:

- Geometry,
- Loading (where compartment load = weight of load units + weight of accessories),
- Profiles used,

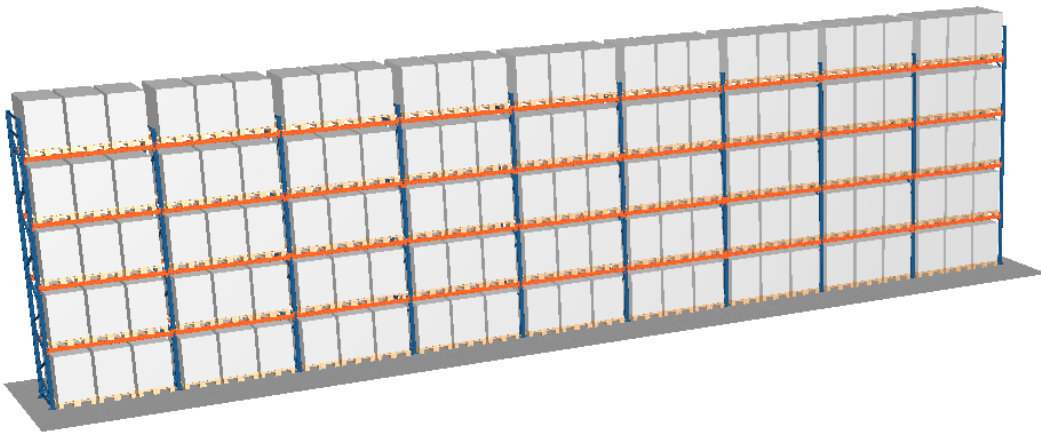


then the existing RDS results are not valid anymore and cannot be therefore displayed.

When the user re-opens the RDS Session Dialogue, a warning message is displayed:

**Row configuration or tag has changed, analysis results have been removed. Please run the RDS analysis again.**

Example – Original structure and RDS Session Dialogue:



**RDS Session Dialog**

RDS\_24008382

**Project** RDS\_24008382

Project modification date 2024-06-28T09:47:09.713Z

Area Area1

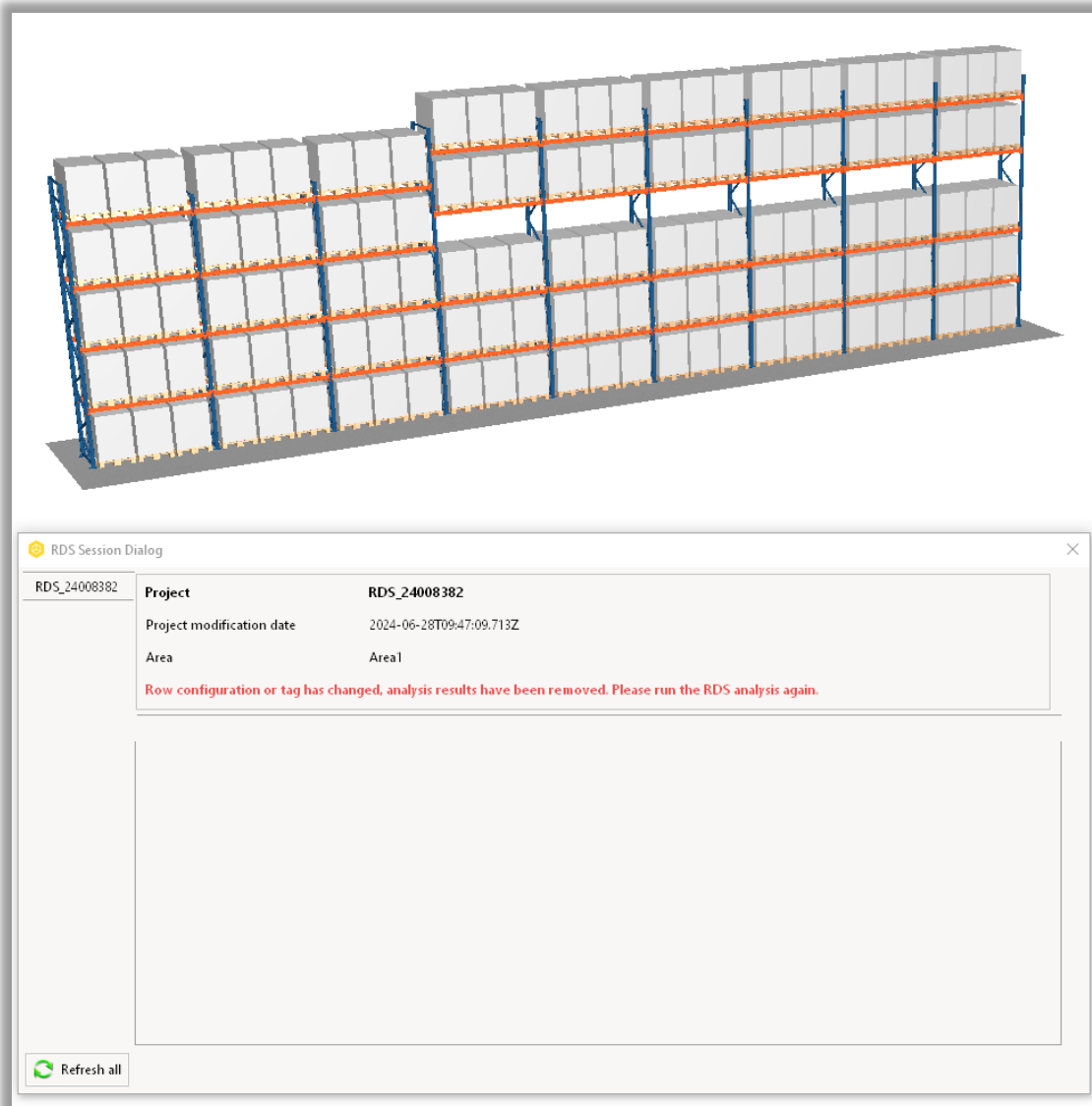
Analysis Analysis result Beam results Downloads

| Cross aisle - Checks on strength |            |        | Down aisle - Checks on strength |            |        |
|----------------------------------|------------|--------|---------------------------------|------------|--------|
| Component                        | UC Section | UC End | Component                       | UC Section | UC End |
| 10068204050PR_M6-S355MC          | 85%        |        | 10068204050PR_M6-S355MC         | 95%        | 18%    |
| U503015_M6-S275JR                | 30%        | 13%    | CC1104015-S275JR                | 99%        | 89%    |

| Cross aisle - Checks on stiffness |     | Down aisle - Checks on stiffness |     |
|-----------------------------------|-----|----------------------------------|-----|
| Deformation                       | UC  | Deformation                      | UC  |
| Upright CA                        | 14% | Upright DA                       | 15% |
|                                   |     | Hook-in beams                    | 99% |

Refresh all

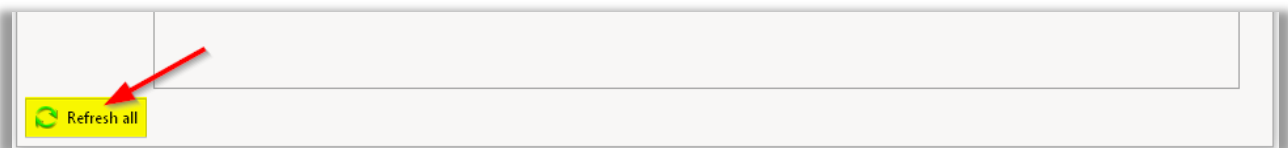
Example - Changed structure and RDS Session Dialogue:



In order to see RDS results again, the user has two options:

**a. Change the structure back**

1. The user needs to change the structure back to its original state,
2. Open the **RDS Session Dialogue**,
3. Click **Refresh all** to fetch the original results from RDS.



**b. Run RDS on the changed structure**

Run the RDS analysis as if analysing a new structure. The already assigned RDS project number stays unchanged and the original results stored in RDS get overwritten.

## 6.2 Copying file with existing RDS results (disconnect from RDS)

The user may decide to copy a CSA file, which has already been linked to RDS project and where RDS results exist:

| CSA file      | RDS project | RDS results valid for |
|---------------|-------------|-----------------------|
| Model_A.cmdrw | RDS_001     | Model A               |

This results in two CSA files linked to the the same RDS project and its results:

| CSA file           | RDS project | RDS results valid for |
|--------------------|-------------|-----------------------|
| Model_A.cmdrw      | RDS_001     | Model A               |
| Model_A_copy.cmdrw | RDS_001     | Model A               |

If the user further modifies the structure in the copied file, it still stays connected to the same RDS project, but the results are not valid for the changed structure anymore:

| CSA file      | RDS project | RDS results valid for |
|---------------|-------------|-----------------------|
| Model_A.cmdrw | RDS_001     | Model A               |
| Model_B.cmdrw | RDS_001     | Model A               |

The user than needs to run RDS analysis again. But if he/she does so, the existing results in RDS are overwritten and the new results are valid just for the changed structure of Model B. They are not valid for the original structure anymore. Such outcome may not be desirable, because results for Model A are lost:

| CSA file      | RDS project | RDS results valid for |
|---------------|-------------|-----------------------|
| Model_A.cmdrw | RDS_001     | Model B               |
| Model_B.cmdrw | RDS_001     | Model B               |

**It is therefore obvious, that it is sometimes desirable to disconnect the CSA file or tag from the already linked project.**

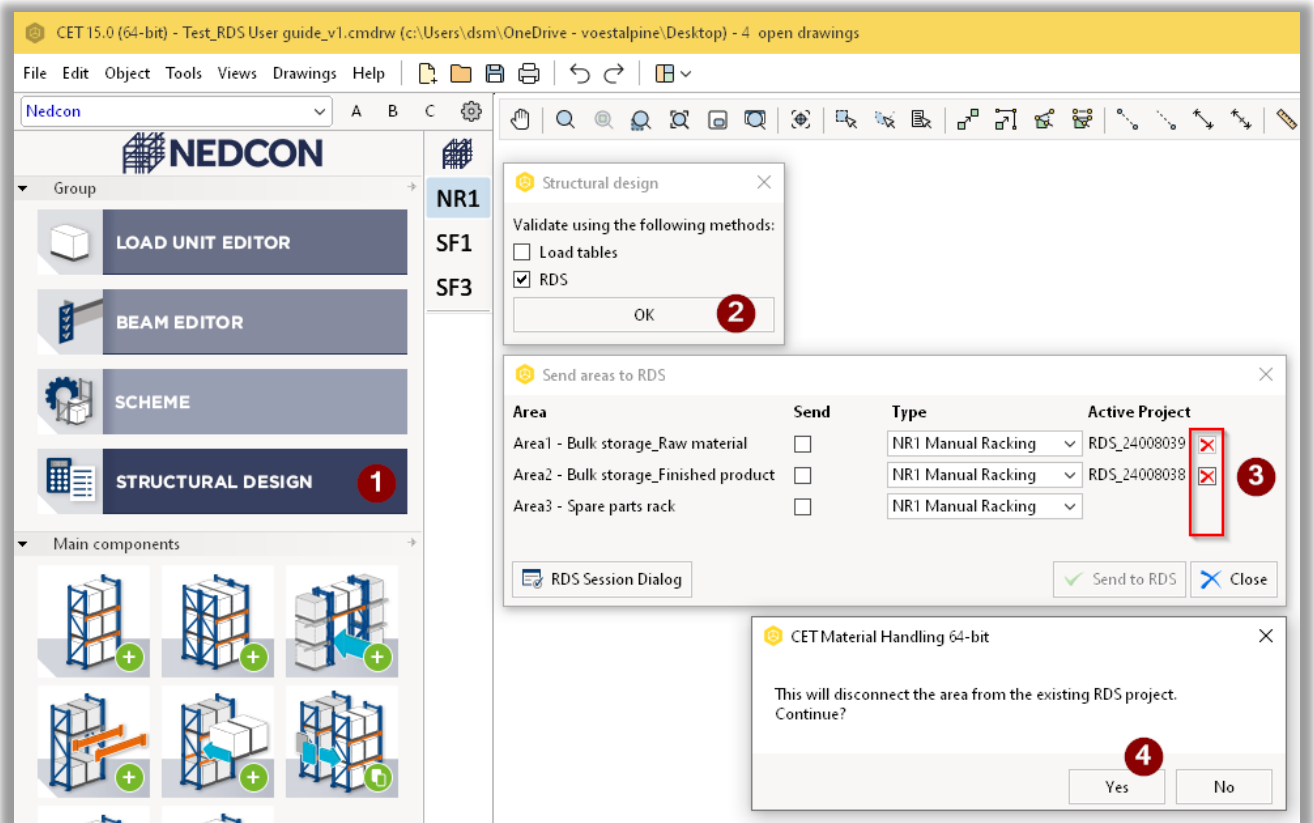


If the user disconnects the Model\_B.cmdrw file from the RDS\_001 project first, i.e. before running RDS analysis on the changed structure, he/she then gets the desirable outcome:

| CSA file      | RDS project | RDS results valid for |
|---------------|-------------|-----------------------|
| Model_A.cmdrw | RDS_001     | Model A               |
| Model_B.cmdrw | RDS_002     | Model B               |

### Disconnecting CSA file or tag from existing RDS project

1. Open dialogue **STRUCTURAL DESIGN** (1),
2. Confirm RDS design method with **OK** (2),
3. Choose all drawing or the tag to be disconnected and click on the corresponding **red cross** (3),
4. Confirm with **Yes** (4).



## 7 Seismic design

### 7.1 General

CSA in the Advanced design mode supports structural engineering design for earthquake resistance of structures for the following countries:

| i  | Country        | Magnitude of Seismic actions       |
|----|----------------|------------------------------------|
| 1  | Austria        | value of $A_{gR}$ [ $m.s^{-2}$ ]   |
| 2  | Belgium        | seismic zone                       |
| 3  | Bulgaria       | value of $A_{gR}$ [ $m.s^{-2}$ ]   |
| 4  | Czech Republic | seismic zone                       |
| 5  | France         | seismic zone                       |
| 6  | Germany        | value of $S_{aP,R}$ [ $m.s^{-2}$ ] |
| 7  | Greece         | seismic zone                       |
| 8  | Norway         | value of $A_{gR}$ [ $m.s^{-2}$ ]   |
| 9  | Romania        | value of $A_{gR}$ [ $m.s^{-2}$ ]   |
| 10 | Slovakia       | value of $A_{gR}$ [ $m.s^{-2}$ ]   |
| 11 | Slovenia       | value of $A_{gR}$ [ $m.s^{-2}$ ]   |
| 12 | Switzerland    | seismic zone                       |

The table above gives also an overview of how magnitude of seismic actions (earthquake) at given location of given country is to be specified:

$A_{gR}$  [ $m.s^{-2}$ ]      reference peak ground acceleration, see EN 1998-1 and EN 1998-1/NA of given country,

$S_{aP,R}$  [ $m.s^{-2}$ ]      spectral response acceleration in the plateau region of the response spectrum,  
where  $S_{aP,R}$  [ $m.s^{-2}$ ] =  $2.5 \cdot A_{gR}$  [ $m.s^{-2}$ ], see DIN EN 1998-1/NA,

seismic zones      see EN 1998-1/NA of given country.

Values of  $A_{gR}$ ,  $S_{aP,R}$ , and seismic zones for given location of given country can be found in the documents listed above or in various sources on internet, e.g.:

<https://www.dlubal.com/en/solutions/online-services/snow-load-wind-speed-and-seismic-load-maps>

Nedcon resellers and employees are legally allowed to use the following credentials on the Dlubal website:

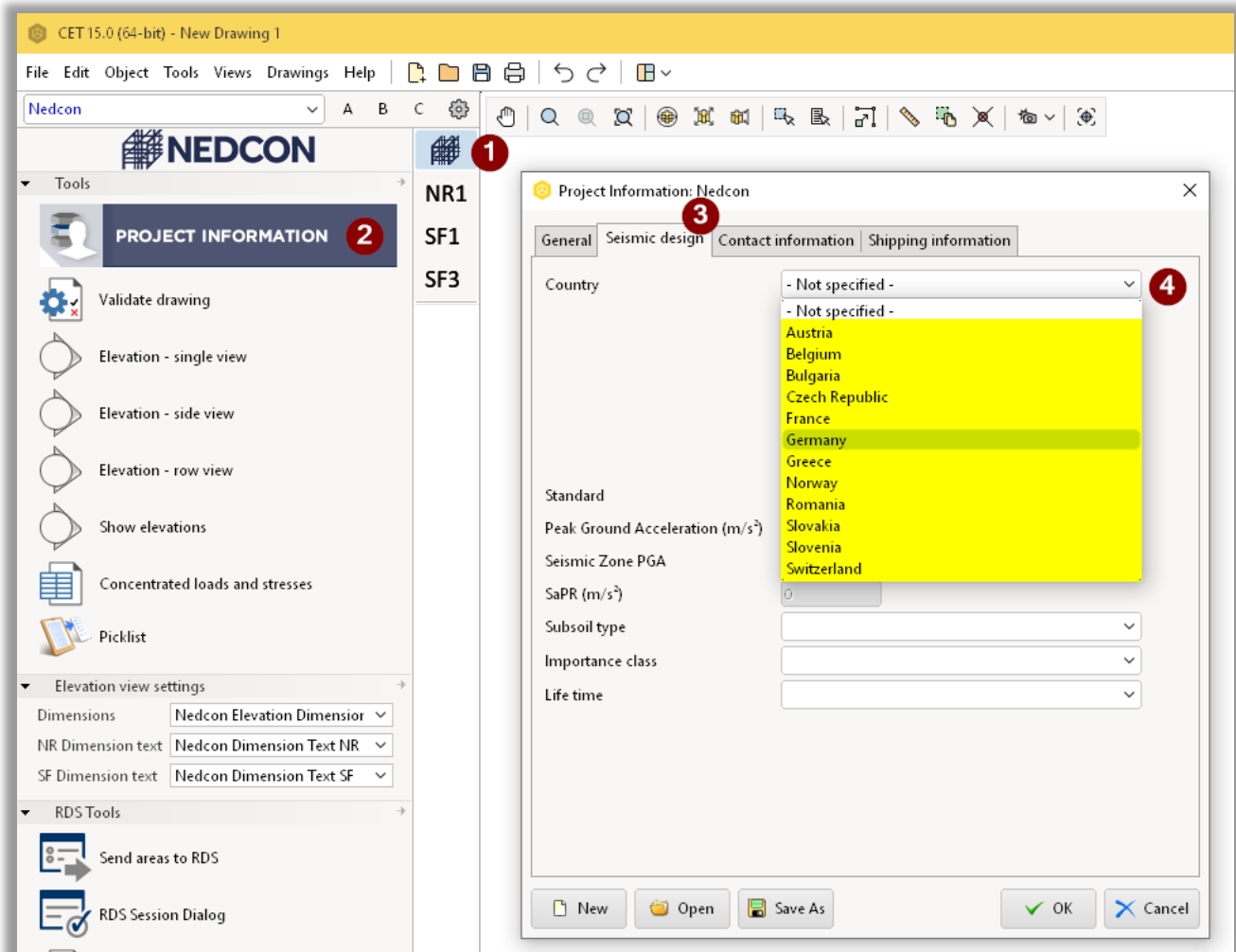
- username      [resellers@nedcon.com](mailto:resellers@nedcon.com)
- password      nedcon

See Annex B for instructions on how to use the Dlubal website to determine seismic parameters. The text is written for RDS and it can be used for CSA.

## 7.2 Activating seismic design

Seismic design is activated by selecting a country to be considered for it:

1. Go to tab **Nedcon** (1) and open **Project Information** (2),
2. Select tab **Seismic design** (3),
3. Using pull-down menu **Country** (4) select the country for seismic design.



## 7.3 Defining seismic parameters

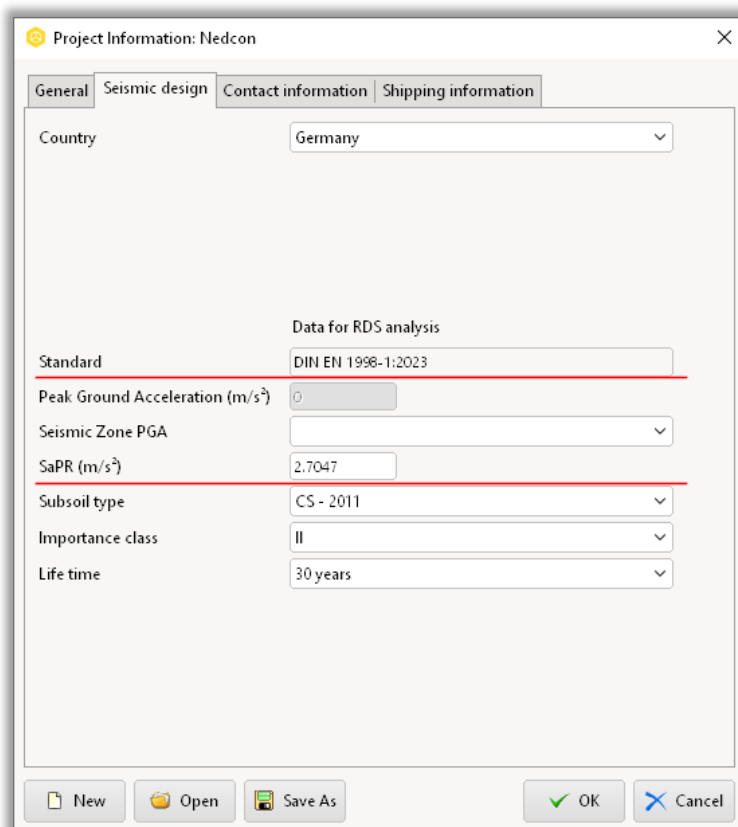
The users define seismic parameters on the tab Seismic design.

### Standard

The standard for seismic design is filled in automatically based on the seismic country selected.

### Peak Ground Acceleration / Seismic Zone (PGA) / SaPR

Magnitude of seismic actions at given location is determined by one of the three parameters depending on the seismic country. The other two parameters are disabled for this country. See section 7.1 and Annex B.



### Subsoil type

Also the Subsoil type options depend on the seismic country selected. See EN 1998-1 and EN 1998-1/NA of the given country.

In case of Germany, the subsoil type consists of:

- Construction subsoil      depth to approx. 30 m      classes A, B, C
- Geological subsoil      depth from approx. 30 m      classes R, T, S

The geological subsoil can be determined using the Dlubal website, see section 7.1 and Annex B.

The construction subsoil shall be determined using local conditions on the construction site.

If the construction subsoil is not known, then it can be conservatively assumed as class C for geological subsoil class R with  $S_{aP,R} \leq 2.0 \text{ m.s}^{-2}$  or for geological subsoil classes T and S for all  $S_{aP,R}$  values.

In case of geological subsoil class R with  $S_{aP,R} > 2.0 \text{ m.s}^{-2}$  the user needs to try both construction subsoil classes B and C and choose the more unfavourable one.

### Importance class

See EN 1998-1 and EN 16681 or National regulations:

#### Class I

- Buildings of minor importance for public safety, e.g. agricultural buildings, etc.
- Warehouses with fully automated storage operations.

#### Class II

- Ordinary buildings, not belonging in the other categories.
- Standard warehouse conditions, including picking areas.

#### Class III

- Buildings whose seismic resistance is of importance in view of the consequences associated with a collapse, e.g. schools, assembly halls, cultural institutions etc.
- Any kind of rack with random public access for private customer.

#### Class IV

- Buildings whose integrity during earthquakes is of vital importance for civil protection, e.g. hospitals, fire stations, power plants, etc.
- Hazardous product storage (subject to the approval of National Authorities); Strategic facilities.

Default value of Importance class for industrial storage structures in CSA is selected as class II. The user is expected to adjust according to the project's conditions.

### Life time (Design life)

See EN 16681 or National regulations:

- |            |                         |                                                 |
|------------|-------------------------|-------------------------------------------------|
| • 30 years | design life of 30 years | valid for Importance classes I and II only (!!) |
| • 50 years | design life of 50 years |                                                 |



Default value of Design life for industrial storage structures in CSA is selected as 30 years. The user is expected to adjust according to the project's conditions.

## 8 CSA projects in RDS application

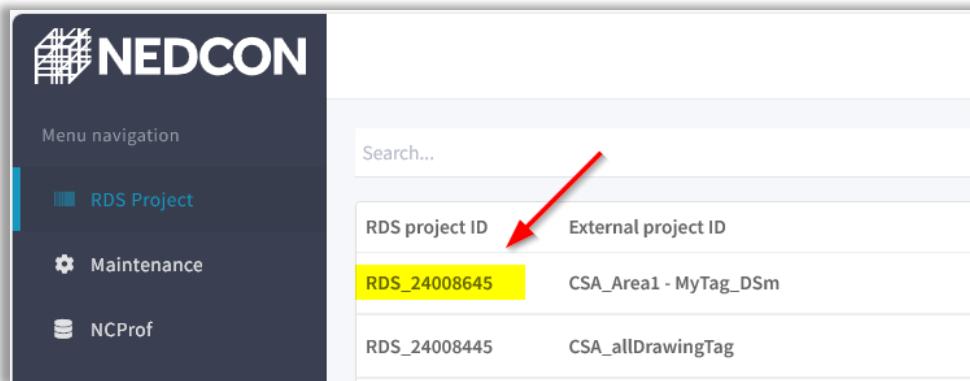
### 8.1 Identification of CSA projects in RDS

It is possible to find and view projects created in CSA directly in the RDS application.

The CSA projects can be identified in RDS with the help of:

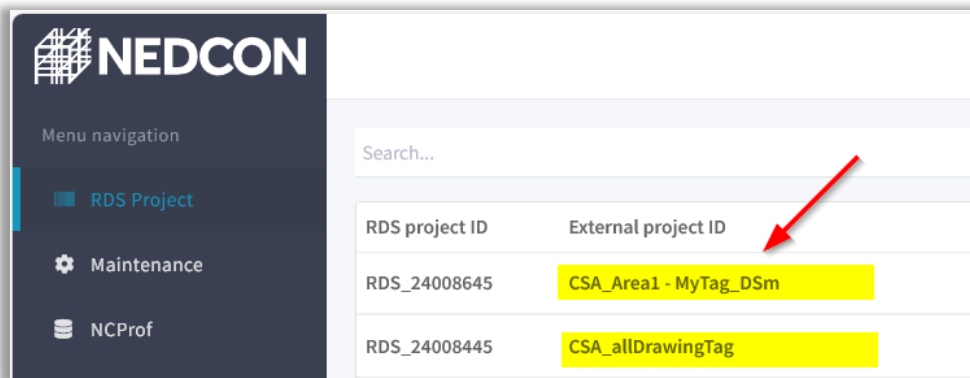
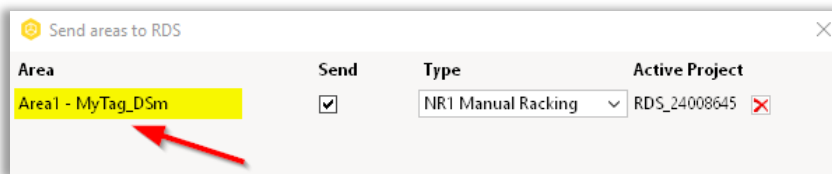
#### a. RDS project ID

The RDS project number automatically assigned to any RDS analysis started from CSA is visible also in RDS in the RDS Project window:



#### b. External project ID

The External project ID of CSA projects starts with a prefix “CSA\_” to which either the structural desing tag name is added or the text “allDrawingTag” in case of the whole drawing being sent to RDS for analysis:



## 8.2 Status of CSA projects in RDS

All CSA projects in RDS are per definition in revision “0” and have a status “Final”:

| RDS project ID | External project ID    | System template | Country | Standard | Status | Revision | Revision date    | State |
|----------------|------------------------|-----------------|---------|----------|--------|----------|------------------|-------|
| RDS_24008645   | CSA_Area1 - MyTag_DSsm | MAN NR1         | Germany | Europe   | Final  | 0        | 16-07-2024 14:21 | ✓     |
| RDS_24008445   | CSA_allDrawingTag      | MAN NR1         | Germany | Europe   | Final  | 0        | 03-07-2024 10:57 | ✓     |

This means, that the projects can be fully viewed in RDS, but they cannot be modified or analysed again:

Project RDS\_24008645 - CSA\_Area1 - MyTag\_DSsm - Revision 0 - Final

### Analysis

This project is final and cannot be analyzed anymore

Project modification date: 16-07-2024 12:21

Latest analysis run: Run number [1]  
Status [Success]  
16-07-2024 14:23

This is arranged on purpose, because there may be just one application, where any changes of a project may be executed. In case of CSA-RDS Integration it is CSA.

If not, then it could happen that the original model in CSA would not correspond to the analysis model in RDS, if changed from within RDS by a user.

## 8.3 Download of RFEM models by Structural Engineers

RDS users having a status of a Structural Engineer in RDS are allowed to download the RFEM models for the given CSA project sent to RDS for analysis:

Last succesful run was 1

Analysis results
Download

Cross aisle RFEM  
Down aisle RFEM  
Post processing Excel  
Report EN  
Report DE

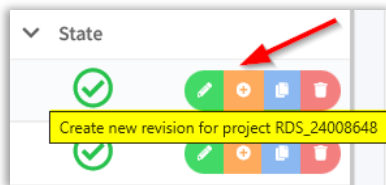
## 8.4 Continue working on CSA projects in RDS

It is possible to continue working on CSA projects in RDS. Any such work stays however limited to RDS and is not transferred back to CSA.

There two ways how to continue in RDS:

### a. RDS - Create a project revision

The RDS user can create a revision of a CSA project:



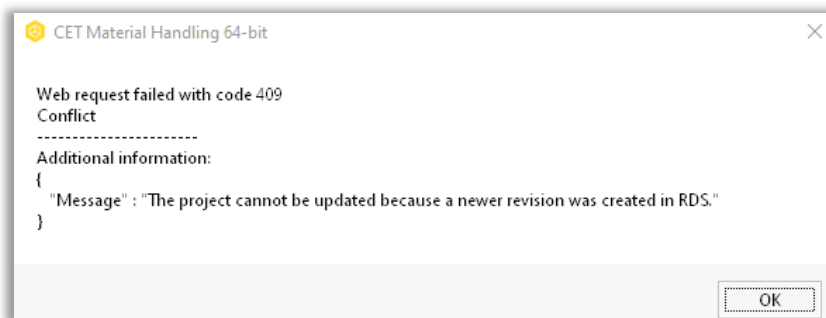
The original revision 0 stays present in RDS and subsequent revisions 1, 2, 3 etc. are created. However, the RDS project number remains unchanged:

| RDS project ID | External project ID   | System template | Country | Standard | Status  | Revision | Revision date    | State |
|----------------|-----------------------|-----------------|---------|----------|---------|----------|------------------|-------|
| RDS_24008648   | CSA_Area1 - MyTag_DSm | MAN NR1         | Germany | Europe   | Concept | 1        | 16-07-2024 17:10 |       |
| RDS_24008648   | CSA_Area1 - MyTag_DSm | MAN NR1         | Germany | Europe   | FINAL   | 0        | 16-07-2024 17:08 |       |

This means that the original project in CSA cannot be sent to RDS anymore, because the RDS project number is already occupied by the higher revision.



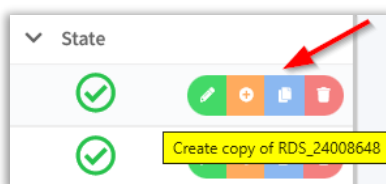
When attempting to run the analysis from CSA, the user then gets the following message:



The CSA user then needs to disconnect the CSA model from the RDS project (see section 6.2) and run the analysis with a new RDS project number.

### b. RDS – Create a project copy

The RDS user can create a copy of a CSA project:



A copy of the project is created in revision 0 and the copy gets a new RDS project number:

| RDS project ID | External project ID          | System template | Country | Standard | Status  | Revision | Revision date    | State |
|----------------|------------------------------|-----------------|---------|----------|---------|----------|------------------|-------|
| RDS_24008649   | Copy of CSA_Area1 - MyTag_DS | MAN NR1         | Germany | Europe   | Concept | 0        | 16-07-2024 17:41 |       |
| RDS_24008648   | CSA_Area1 - MyTag_DS         | MAN NR1         | Germany | Europe   | Final   | 0        | 16-07-2024 17:08 | ✓     |

This means that the original project in CSA can still be sent to RDS,  
because the RDS project number is not occupied by the copy.



Creating a copy of a CSA project in RDS is therefore preferred for subsequent work in RDS over creating a revision.

## **Annex A – Limitations of RDS design method**

## A1. General

When operating in the Advanced design mode then the RDS design method can be used for any pallet racking structure modelled in CSA with the exceptions and limitations listed below.

## A2. Number of beam levels

Maximum number of beam levels supported by RDS design is 15 beam levels.



## A3. One upright frame configuration per RDS run

RDS currently supports only one upright frame configuration per RDS analysis. The upright frame configuration consists of, where applicable:

- Upright profile, (from 08067204050PR to 12083255070PR),
- Upright steel grade, (S355MC for S350GD),
- Frame bracing profile (U or OB),
- Frame bracing steel grade (S275JR or S250GDS),
- Frame arrangement (450-2 or 600-2).

This means that there has to be only:

- one upright frame configuration per drawing, if all drawing is sent to RDS, or
- one upright frame configuration per tag, if STRUCTURAL DESIGN tags are used.



This limitation will be lifted in the near future and RDS design method will support optional varying upright frame configurations for selected bays, e.g. for upright frames of Tunnel bays.

## Frame heights in CSA and in RDS

Note, that frame height is not part of the upright frame configuration. The upright frames in CSA may have varying frame heights when using RDS design. The tallest (highest) intermediate upright frame is sent to RDS for analysis.



## A4. Custom beam profiles

Any custom beam profile can be created using the Beam Editor except for CC090/NC090 beams combined with the 7HK hook-in connector. This configuration is not defined and is not supported.

List of possible custom beam profiles:

|                |         |         |         |   |
|----------------|---------|---------|---------|---|
| CC0904015A_275 | PR2-4HK | PR2-5HK | N/A     | ← |
| CC1104015A_275 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| CC1105015A_275 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| CC1304015A_275 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| CC1305015A_275 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| CC1505015A_355 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| NC0904015A_275 | PR2-4HK | PR2-5HK | N/A     | ← |
| NC1104015A_275 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| NC1304015A_275 | PR2-4HK | PR2-5HK | PR2-7HK |   |
| NC1505015A_355 | PR2-4HK | PR2-5HK | PR2-7HK |   |



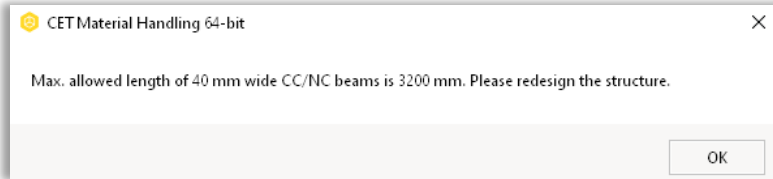
## A5. Upright - beam combinations

Any beam can be connected to any upright except for beams with 5HK and 7HK hook-in connectors which cannot be connected to 80 mm wide uprights (08067204050PR). They do not fit.



## A6. Beam length

CC/NC beams 40 mm wide are not allowed to be longer than 3200 mm. The length is limited to ensure safe interaction with servicing equipment and to reduce lateral-torsional buckling.



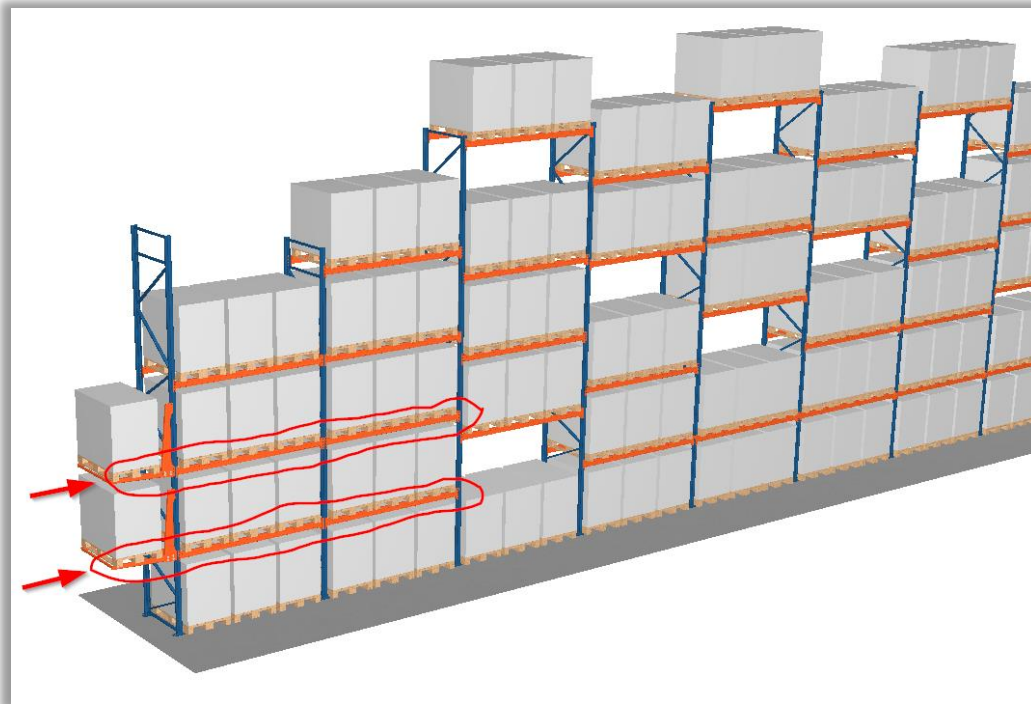
## A7. Rules for attaching P&D stations

The rack structure has to comply with the following rules before a P&D station can be attached in the Advanced design mode:

- min. 2 beam levels,
- min. 3 bays,
- max. 1250 kg of load unit weight on a P&D station.



Further, in order to be able to attach a P&D station to a particular beam level, the beams in the adjoining first two bays has to be aligned (be on the same height level). The rule is designed to reduce fatigue loading on hook-in connectors and other components.



## A8. Anchoring and Baseplate checks

RDS currently does not check anchoring for tension and shear. CSA therefore also cannot check the anchoring. The verifications shall be performed outside of the applications.



RDS further currently does not list checks of baseplate in compression and tension in the “Summary of Post Processing Results” dialogue. CSA therefore cannot list the baseplate checks in the “RDS Session Dialogue”. The baseplate checks are however available in the RDS reports in pdf and in the Postprocessing file in Excel (see section 2.10).



Both Anchoring and Baseplate checks will be added to the applications in the near future.

## A9. RDS General remarks

The following remarks are attached to RDS reports in pdf by RDS. They are valid for all analyses performed by RDS:



1. For the frame arrangement see the frame arrangement report.
2. If components are not listed in this report, then they are not checked by RDS.
3. In case of product load eccentricities in frame direction, only additional upright loading is taken into account.  
No other effects have been considered. In case of eccentricity factor larger than 0.6, consult a structural engineer
4. No strength reduction is applied over beams loaded with point loads.
5. RDS has some limitations which can influence the results, see the RDS manual for more information.
6. General remark: Always consult a structural engineer for final approval and release of a project!

## **Annex B – Obtaining local seismic parameters for RDS (by Mark Assink)**

## Manual for seismic properties in RDS

This manual describes how to obtain local seismic parameters as input for RDS seismic analysis.

The seismic parameters can be found with the website shown in the link below:

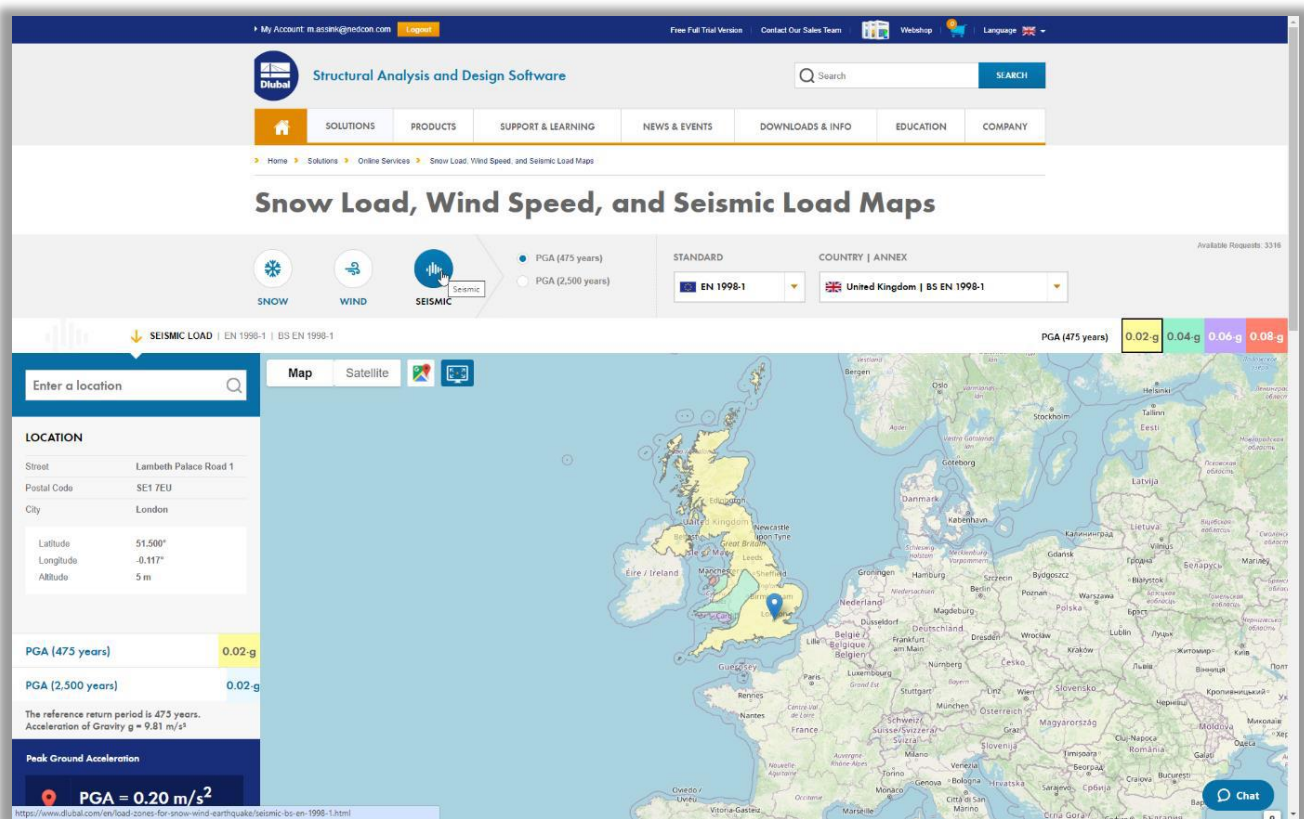
<https://www.dlupal.com/en/solutions/online-services/snow-load-wind-speed-and-seismic-load-maps>

Make sure you are logged in at the Dlubal site (see the top blue part of the Dlubal website):

- Username: [resellers@nedcon.com](mailto:resellers@nedcon.com)
- Password: nedcon

Below an example how to obtain German seismic properties is shown. For all other countries, the method is comparable.

Click on button “Seismic”:



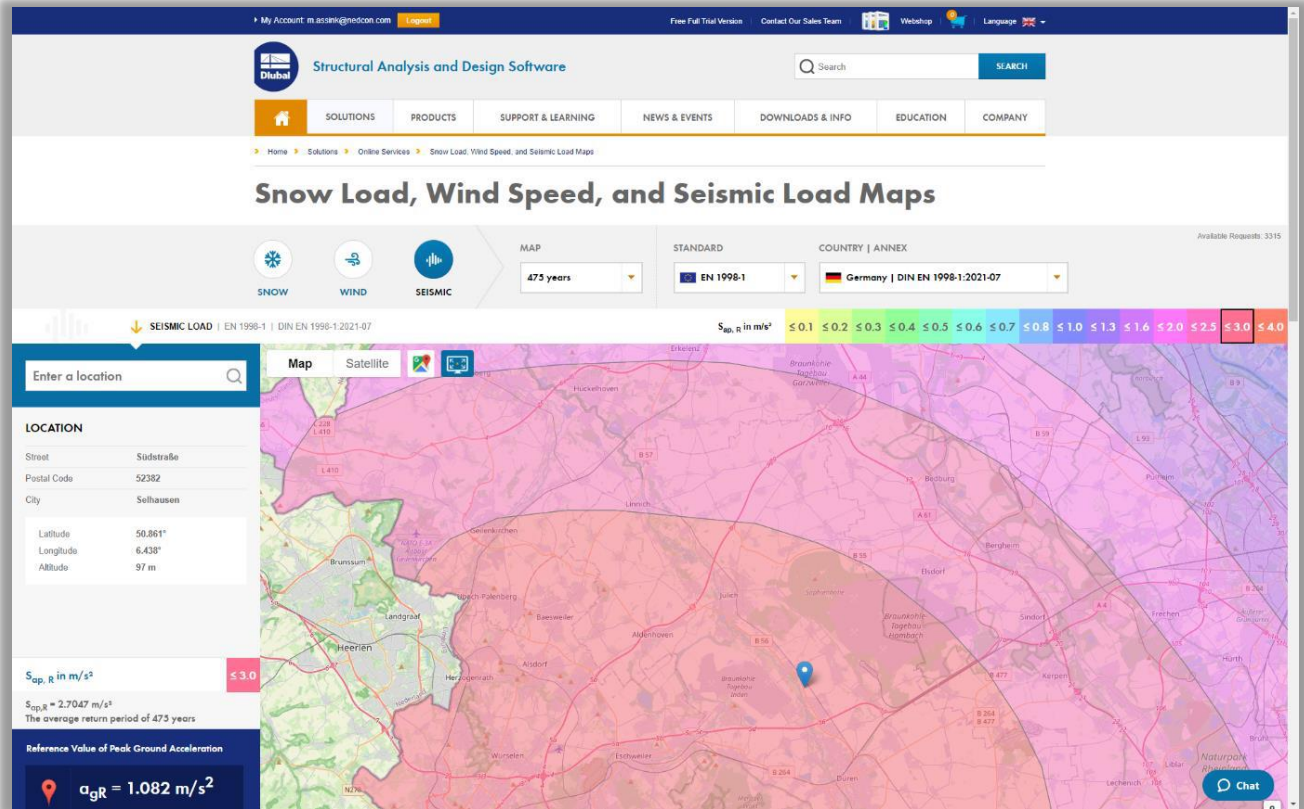
Select the correct standard:

- For Germany: “EN 1998-1”
- Select “Germany | DIN EN 1998-1:2021-07” as “Country”

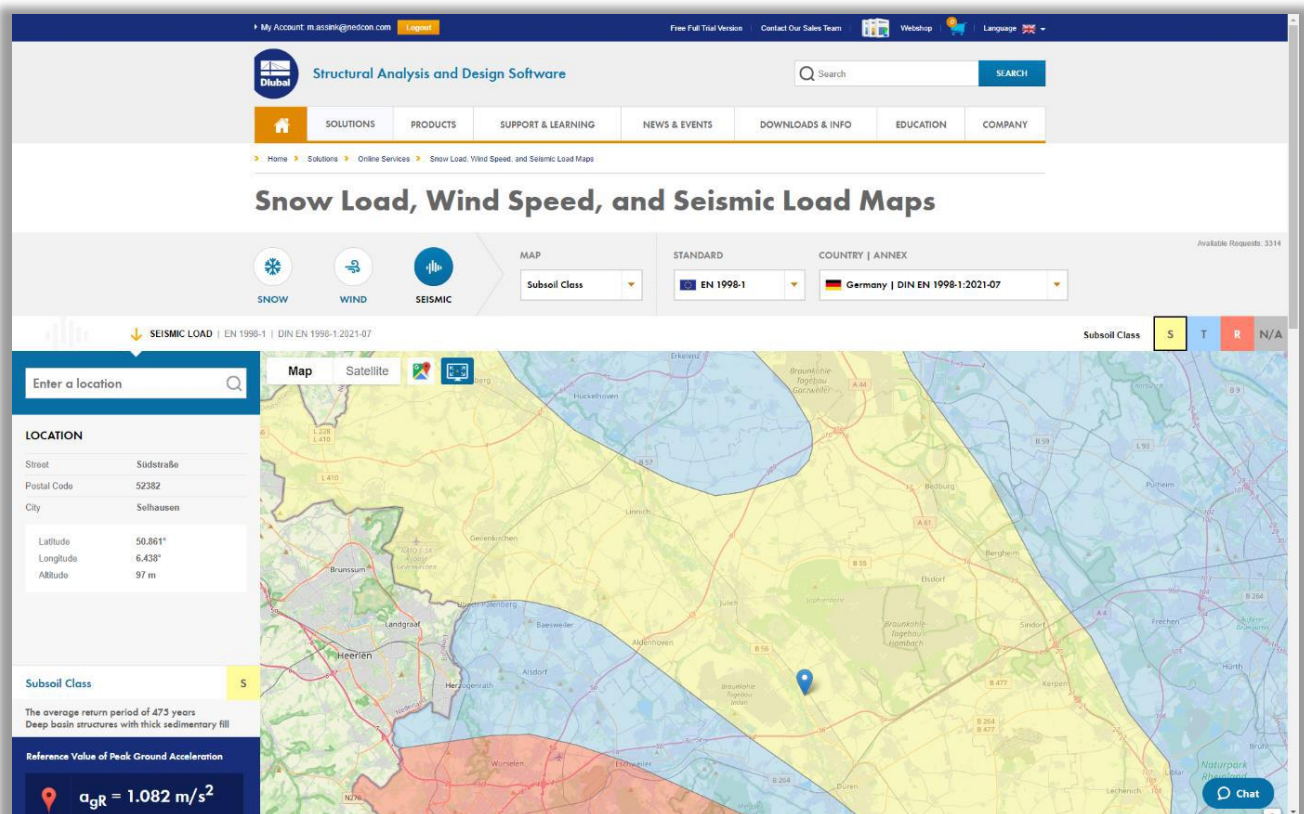
Choose the correct location by:

- Zoom on the map and click on the building site location
- Enter a location at the text box “Enter a location” on the website

Read the given values from the website in the left part of the screen. In the example this is a  $S_{apR}$  value of  $2.7047 \text{ m/s}^2$ , see the figure below. Use this value in RDS.



Specifically for Germany, it is also possible to select a soil class. You have to select “Subsoil class” as “Map” and read the sub soil class value on this location. In this example this is “S”:



The values of above shall be used as input in RDS at the seismic analysis tab. Do not change any other RDS default values, unless you know what you are doing.

For the majority of the European countries, RDS provides support for seismic analysis.

For all the countries listed below, the map “EN 1998-1” shall be selected:

- Austria ÖNORM B 1998-1
  - Select map: “ $a_{gR}$  in  $m/s^2$ ”
- Belgium NBN EN 1998-1
- Bulgaria BDS EN 1998-1
  - Select map: “475 years”
- Czech Republic ČSN EN 1998-1
- France NF EN 1998-1
- Germany DIN EN 1998-1:2021-07 or 2023-11
  - Select map: “475 years”
  - Select map: “Subsoil class”
- Greece ΕΔΟΤ EN 1998-1
- Norway NS-EN 1998-1
- Romania SR EN 1998-1
  - Select map: “ $a_{gR}$  in  $m/s^2$ ”
- Slovakia STN EN 1998-1
- Slovenia SIST EN 1998-1:2009-01

For Switzerland, you should select standard “SIA 261”:

- Switzerland - standard “SIA 261”
  - Select map: “Seismic zone”